

**Directions :** For question number 32, two statements are given, one labelled as **Assertion (A)** and the other is labelled as **Reason (R)**. Select the correct answer to this question from the codes (A), (B), (C) and (D) as given below :

- (A) Both, Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
- (B) Both, Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
- (C) Assertion (A) is true, but Reason (R) is false.
- (D) Assertion (A) is false, but Reason (R) is true.

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**32. Assertion (A) :** When rays of white light pass through a prism, on emerging they give spectrum of seven colours.

**Reason (R) :** It is due to the scattering of light that red light bends minimum and violet light bends the maximum.



**Directions :** For question number 32, two statements are given, one labelled as **Assertion (A)** and the other is labelled as **Reason (R)**. Select the correct answer to this question from the codes (A), (B), (C) and (D) as given below :

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**Reason (R) :** It is due to the scattering of light that red light bends minimum and violet light bends the maximum.

1

(C) / Assertion (A) is true, but Reason (R) is false.



**31.** A thin beam of white light falls on a face of a glass prism and a spectrum is obtained on a screen. Which of the following options is the correct sequence of colours from top to bottom on the screen ?

*1*

- (A) Red, Green, Yellow, Violet, Blue
- (B) Red, Yellow, Green, Blue, Violet
- (C) Violet, Blue, Green, Yellow, Red
- (D) Violet, Red, Green, Yellow, Blue



**31.** A thin beam of white light falls on a face of a glass prism and a spectrum is obtained on a screen. Which of the following options is the correct sequence of colours from top to bottom on the screen ? 1

- (A) Red, Green, Yellow, Violet, Blue
- (B) Red, Yellow, Green, Blue, Violet
- (C) Violet, Blue, Green, Yellow, Red
- (D) Violet, Red, Green, Yellow, Blue

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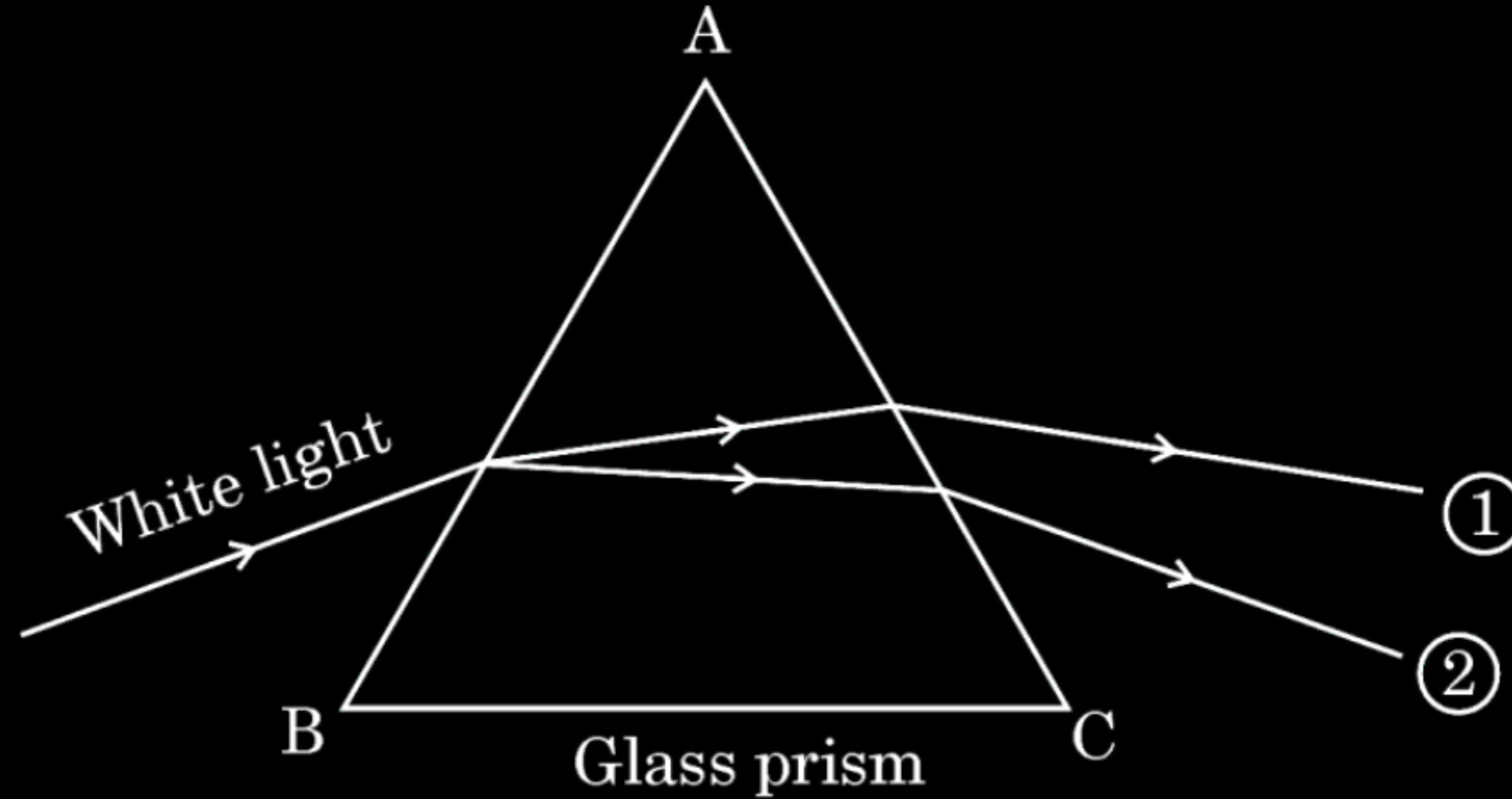
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(B) / Red, Yellow, Green, Blue, Violet



31. Observe the given ray diagram showing refraction through a prism and choose the correct option for colour of rays 1 and 2 :

1

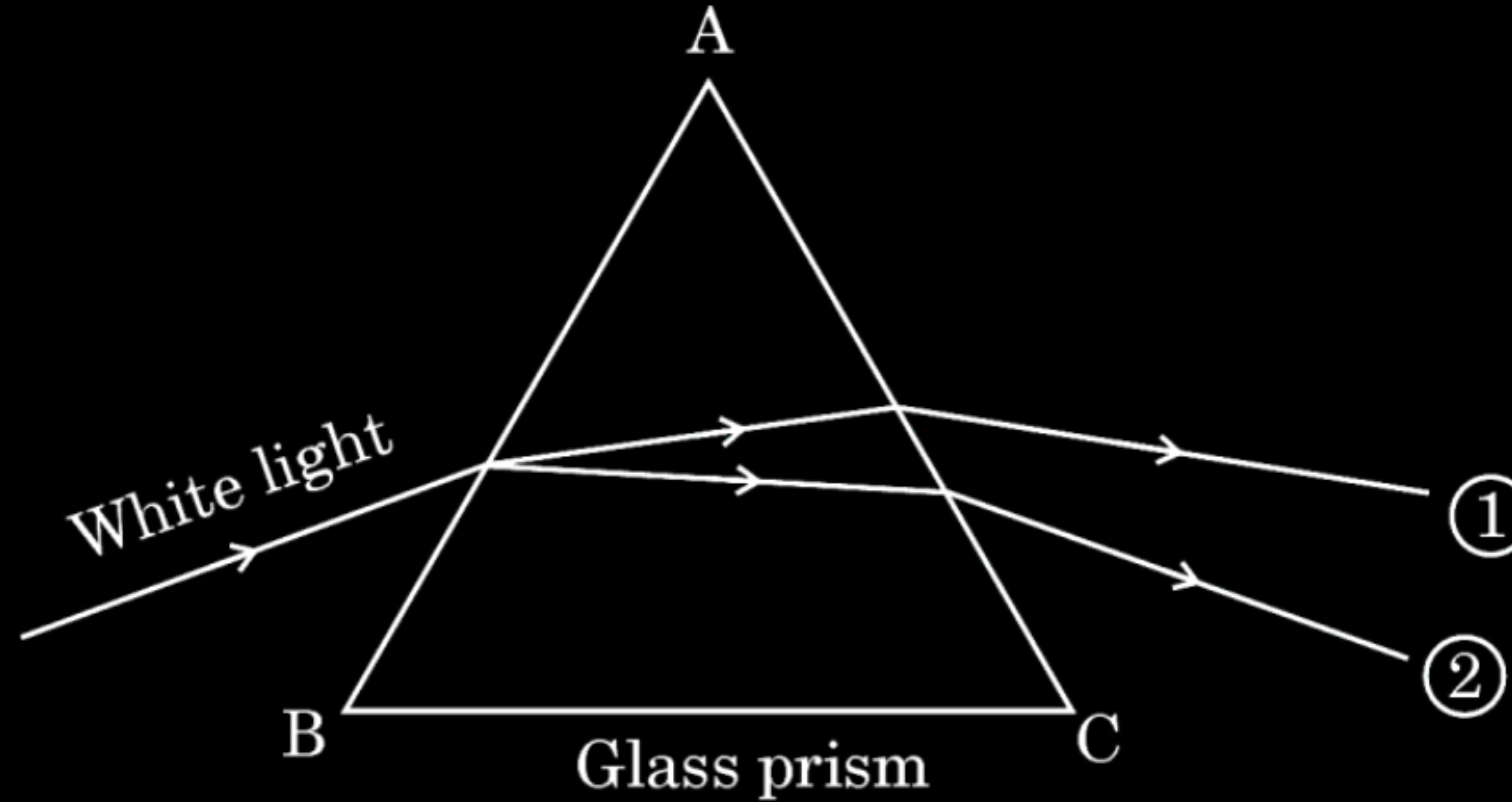


|     | <i>Colour of ray 1</i> | <i>Colour of ray 2</i> |
|-----|------------------------|------------------------|
| (A) | Blue                   | Red                    |
| (B) | Green                  | Yellow                 |
| (C) | Red                    | Violet                 |
| (D) | Violet                 | Red                    |



31. Observe the given ray diagram showing refraction through a prism and choose the correct option for colour of rays 1 and 2 :

1



|     | <i>Colour of ray 1</i> | <i>Colour of ray 2</i> |
|-----|------------------------|------------------------|
| (A) | Blue                   | Red                    |
| (B) | Green                  | Yellow                 |
| (C) | Red                    | Violet                 |
| (D) | Violet                 | Red                    |

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(C) /Colour of ray 1- Red; Colour of ray 2 - Violet



- (b) With the help of a ray diagram, explain any phenomenon of formation of spectrum in nature.

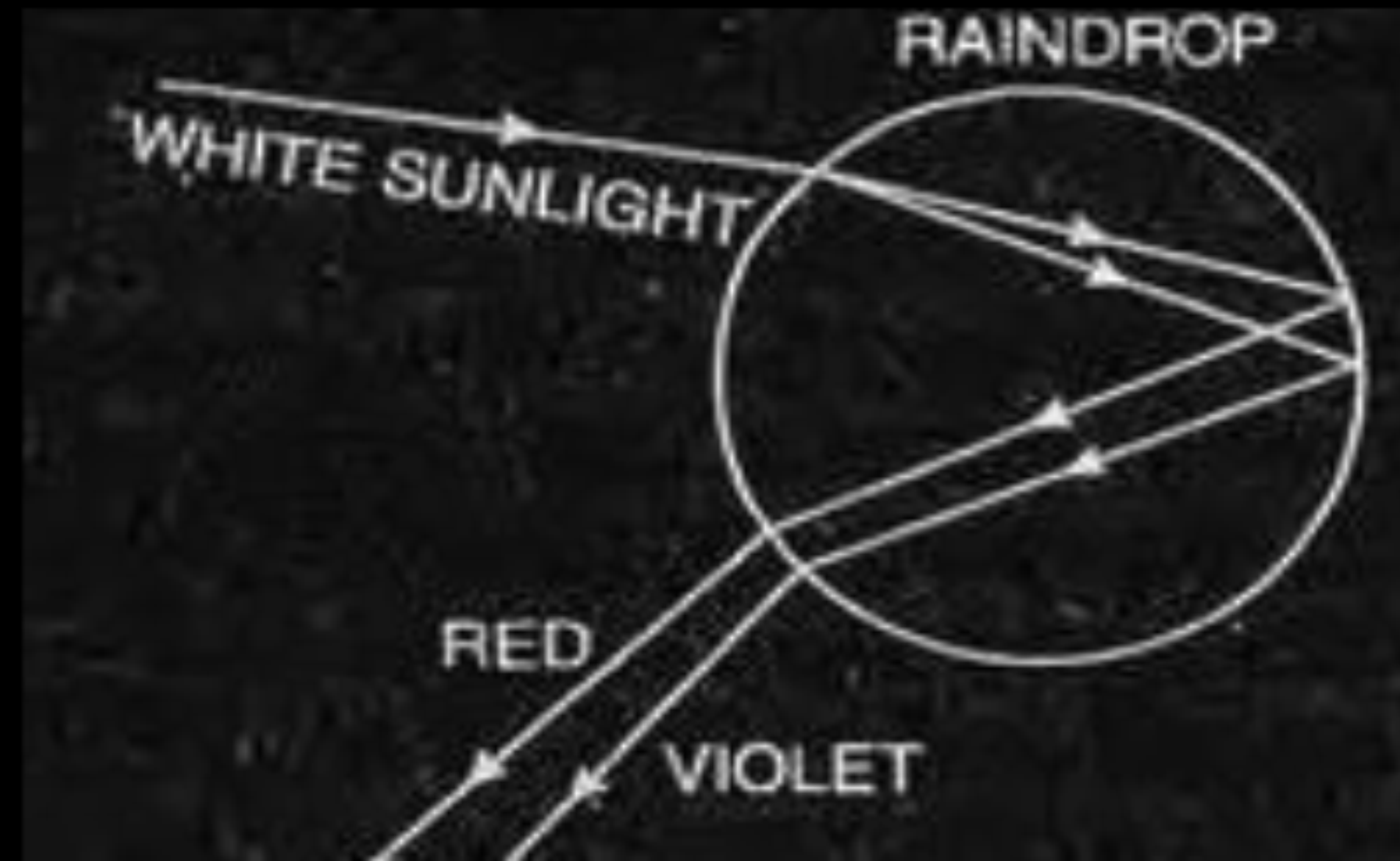


- (b) With the help of a ray diagram, explain any phenomenon of formation of spectrum in nature.

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2

(b)



A natural spectrum like rainbow is formed when sunlight passes through tiny water droplets present in the atmosphere. The water droplets disperse the incident light ray, then reflect it internally and finally refract it again when it comes out of the raindrop.

(any other phenomenon of spectrum in nature)

1

1



**31.** Angle of deviation in refraction of light through a glass prism is the :

1

- (A) Angle between the normal at the face on the emerging side of the prism and the emergent ray.
- (B) Angle between the normal at the face on the emerging side of the prism and the incident ray.
- (C) Angle between the incident ray and the refracted ray.
- (D) Angle between the directions of incident ray and emergent ray.



**31.** Angle of deviation in refraction of light through a glass prism is the :

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- (A) Angle between the normal at the face on the emerging side of the prism and the emergent ray.
- (B) Angle between the normal at the face on the emerging side of the prism and the incident ray.
- (C) Angle between the incident ray and the refracted ray.
- (D) Angle between the directions of incident ray and emergent ray.

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(D) /Angle between the directions of incident ray and emergent ray.

---



- (a) Name the phenomenon due to which a prism splits the incident white light into a band of colours. State the reason of the phenomenon also.



(a) Name the phenomenon due to which a prism splits the incident white light into a band of colours. State the reason of the phenomenon also.

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2

|                                                                                                                    |   |
|--------------------------------------------------------------------------------------------------------------------|---|
| (a)                                                                                                                |   |
| • Dispersion of light                                                                                              | 1 |
| • Different colours of light bend through different angle with respect to incident ray as they pass through prism. | 1 |



35. (a) Draw the ray diagram for refraction of light through a glass prism and mark angle of refraction and angle of deviation.
- (b) When the path of a light ray refracted through a glass prism is reversed how will the angle of deviation change ? Explain.

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35. (a) Draw the ray diagram for refraction of light through a glass prism and mark angle of refraction and angle of deviation.
- (b) When the path of a light ray refracted through a glass prism is reversed how will the angle of deviation change ? Explain.

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3

(a)

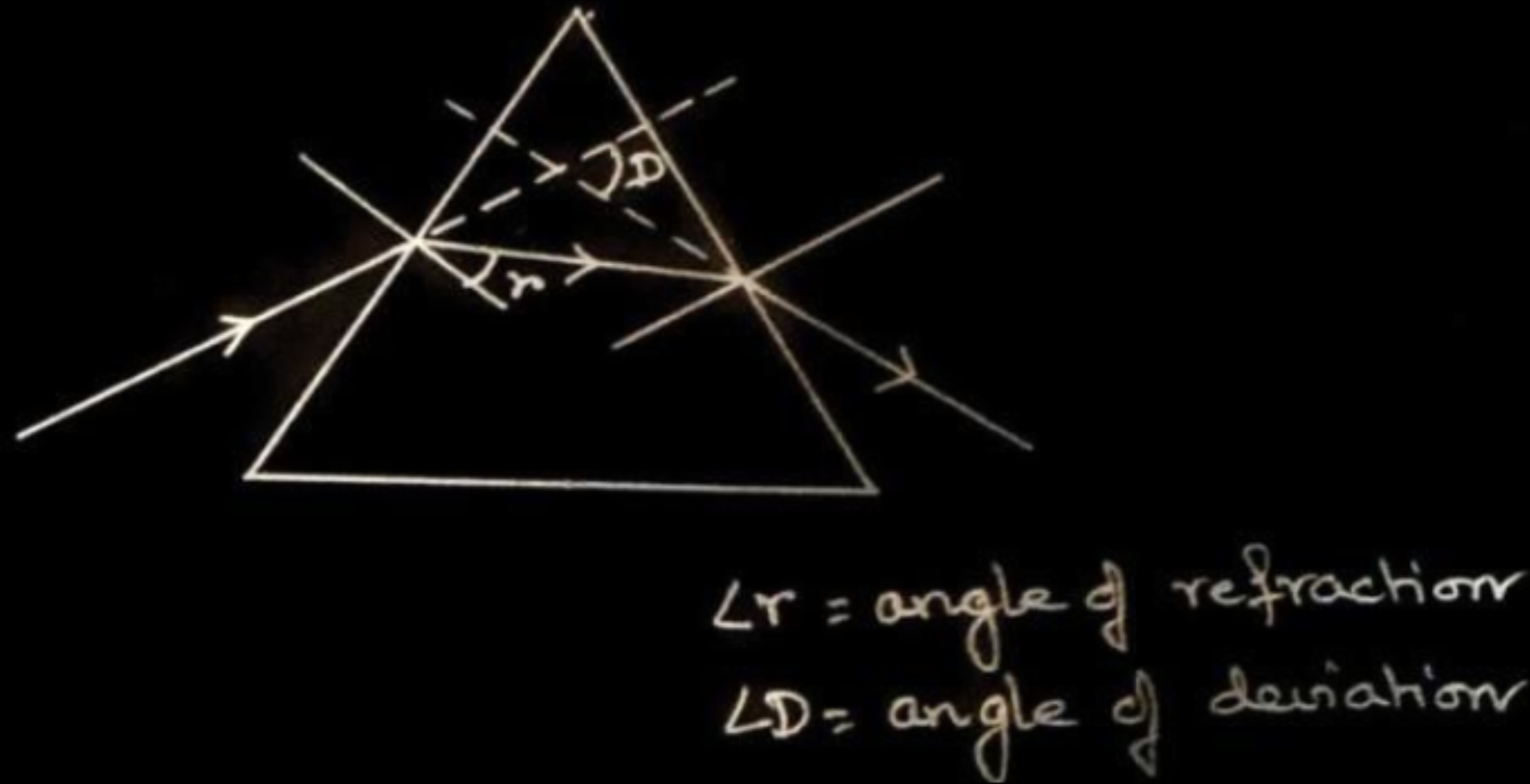


Diagram  
Labelling

**1**  
 $\frac{1}{2} + \frac{1}{2}$

(b)

- Angle of deviation will not change.
- The ray of light will retrace its original path.

$\frac{1}{2}$   
 $\frac{1}{2}$



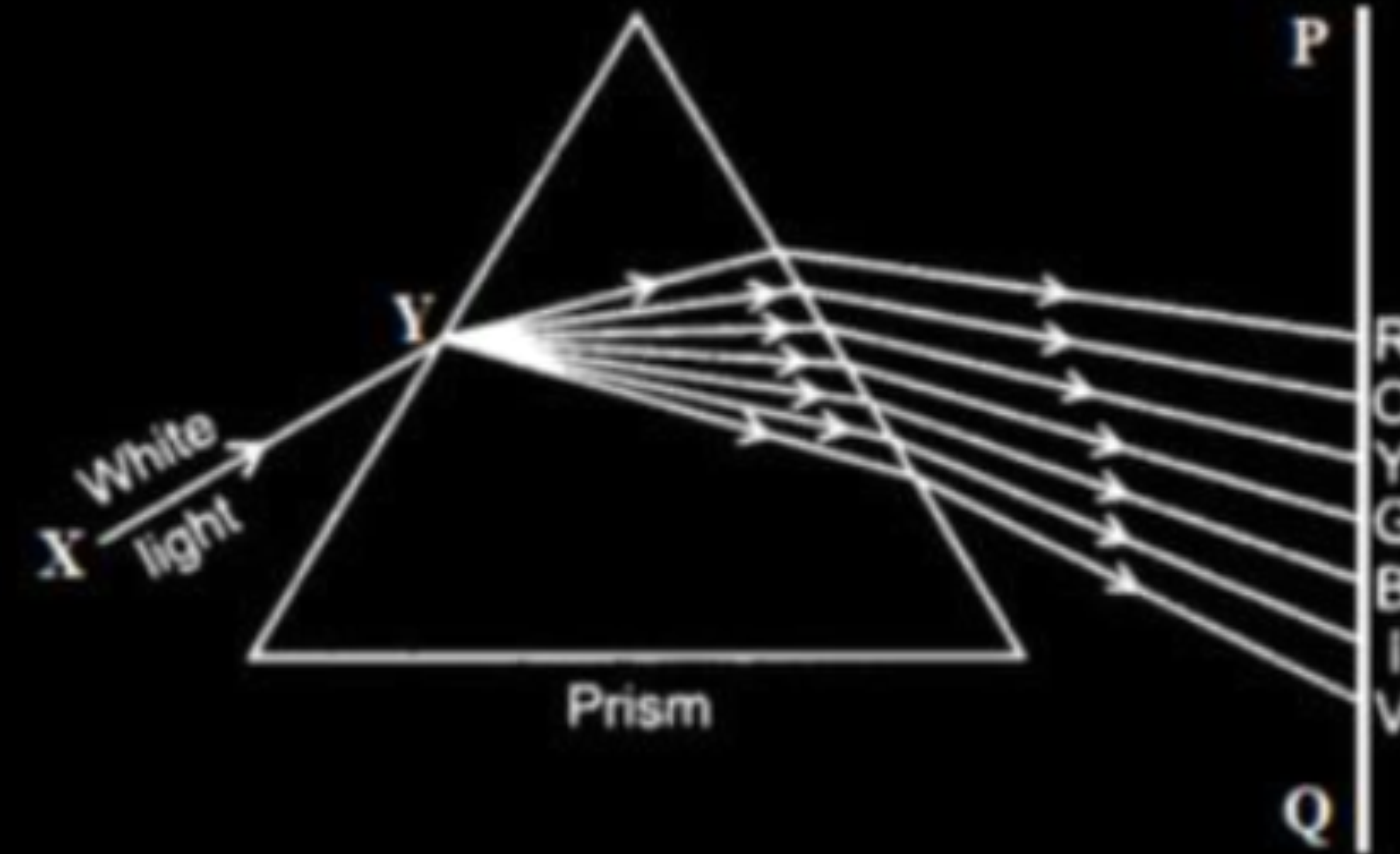
**33.** Draw a labelled diagram to show (i) dispersion of a beam of white light, and (ii) formation of a rainbow.



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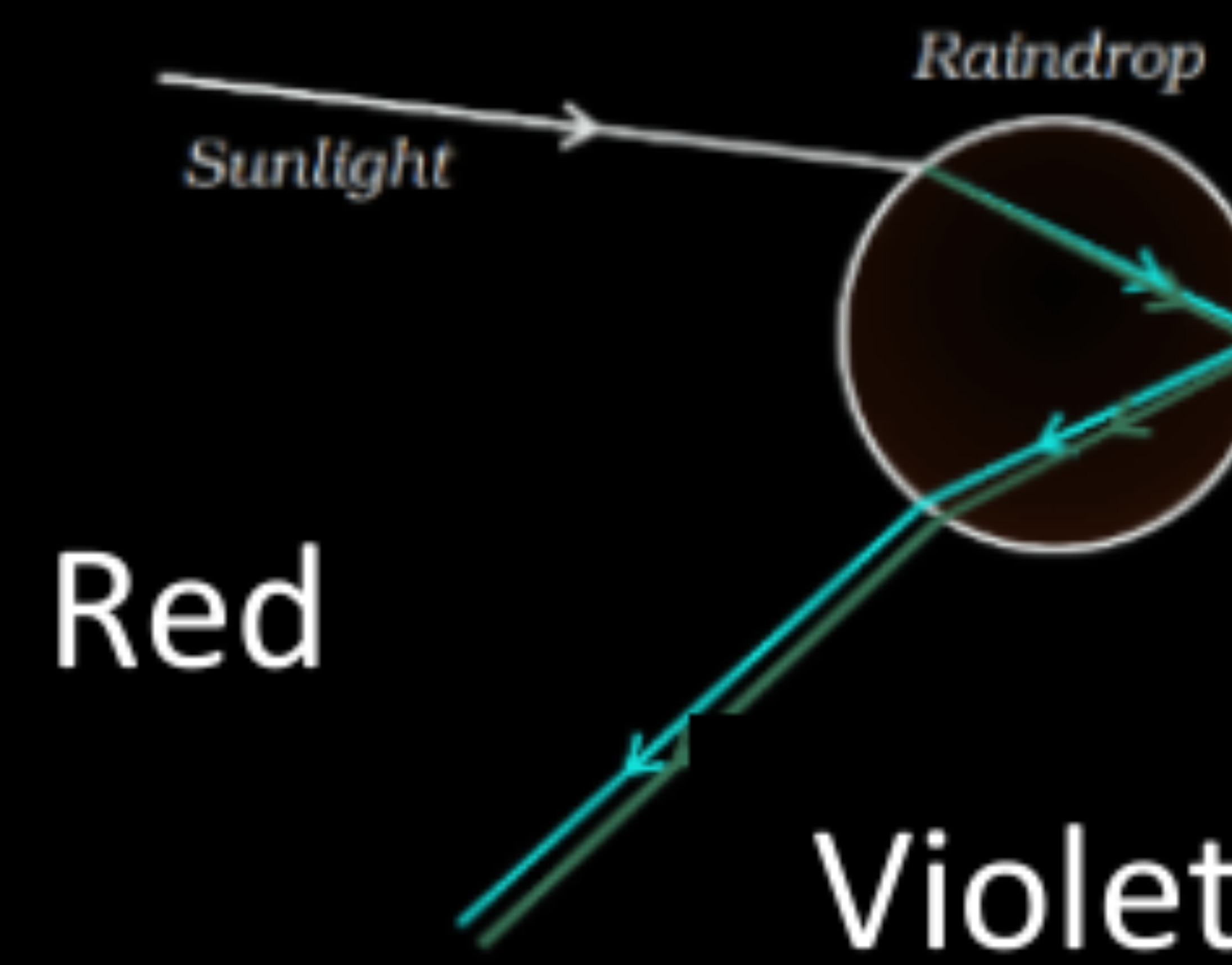
3

(i)



1½

(ii)



1½



**19.**    *Assertion (A) :* The angle through which a ray of light bends on passing through a prism is called the angle of deviation.

*Reason (R) :*    The peculiar shape of a prism makes the emergent ray bend at an angle to the direction of the incident ray.



**19.**    *Assertion (A) :* The angle through which a ray of light bends on passing through a prism is called the angle of deviation.

*Reason (R) :*    The peculiar shape of a prism makes the emergent ray bend at an angle to the direction of the incident ray.

(a)    Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).



- 25.** (a) When and where does a rainbow appear in the sky ? Draw a labelled ray diagram to show its formation. 2

**OR**

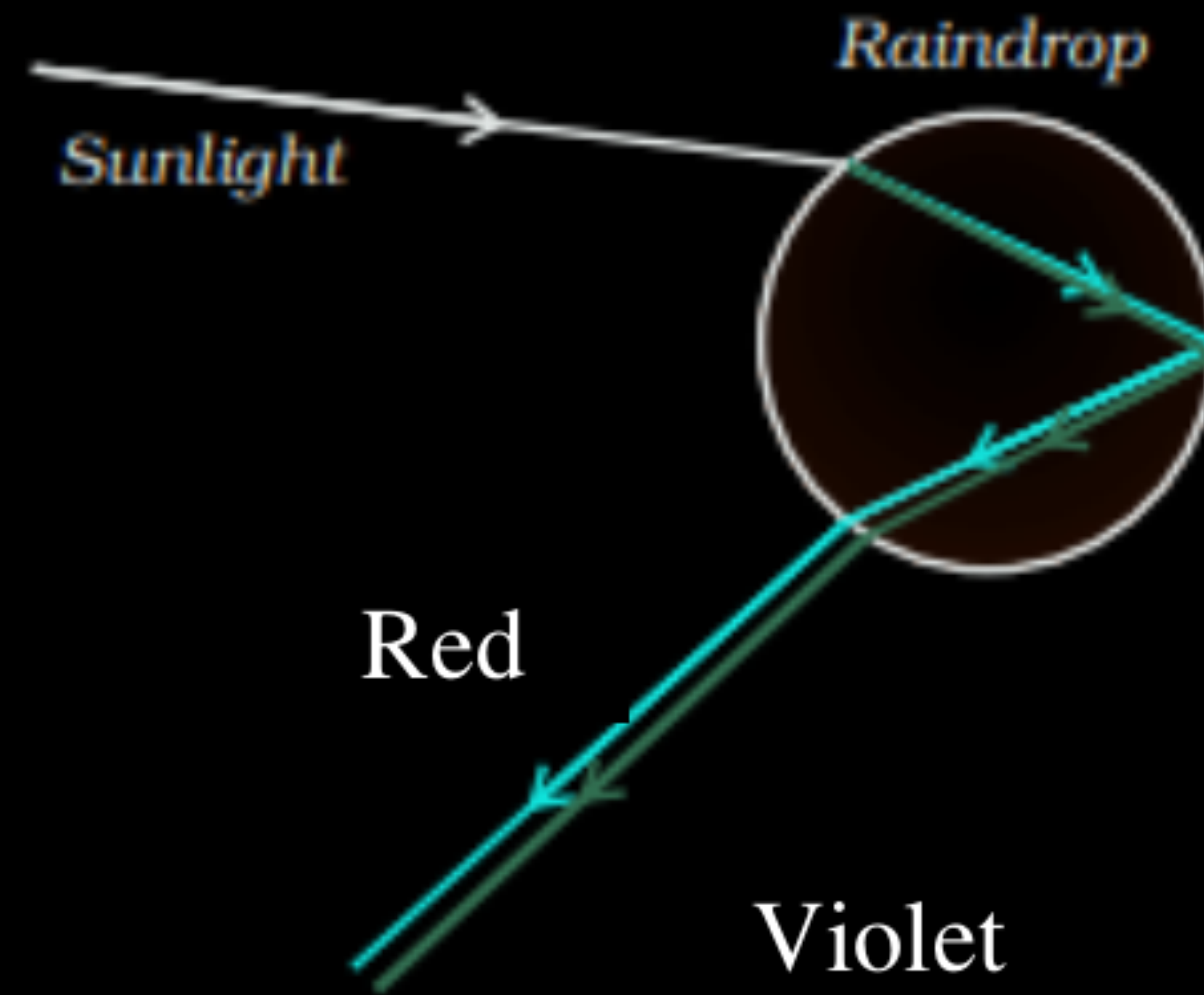
- (b) What is scattering of light ? Why does the clear sky appear blue ? 2



- (a)
- It is formed after a rain shower.
  - It is always formed in a direction opposite to that of the Sun.

$\frac{1}{2}$

$\frac{1}{2}$



1

**OR**

- (b) (i) The phenomenon of the change in direction of propagation of light caused by large sized molecules/ caused by colloidal particles.

1

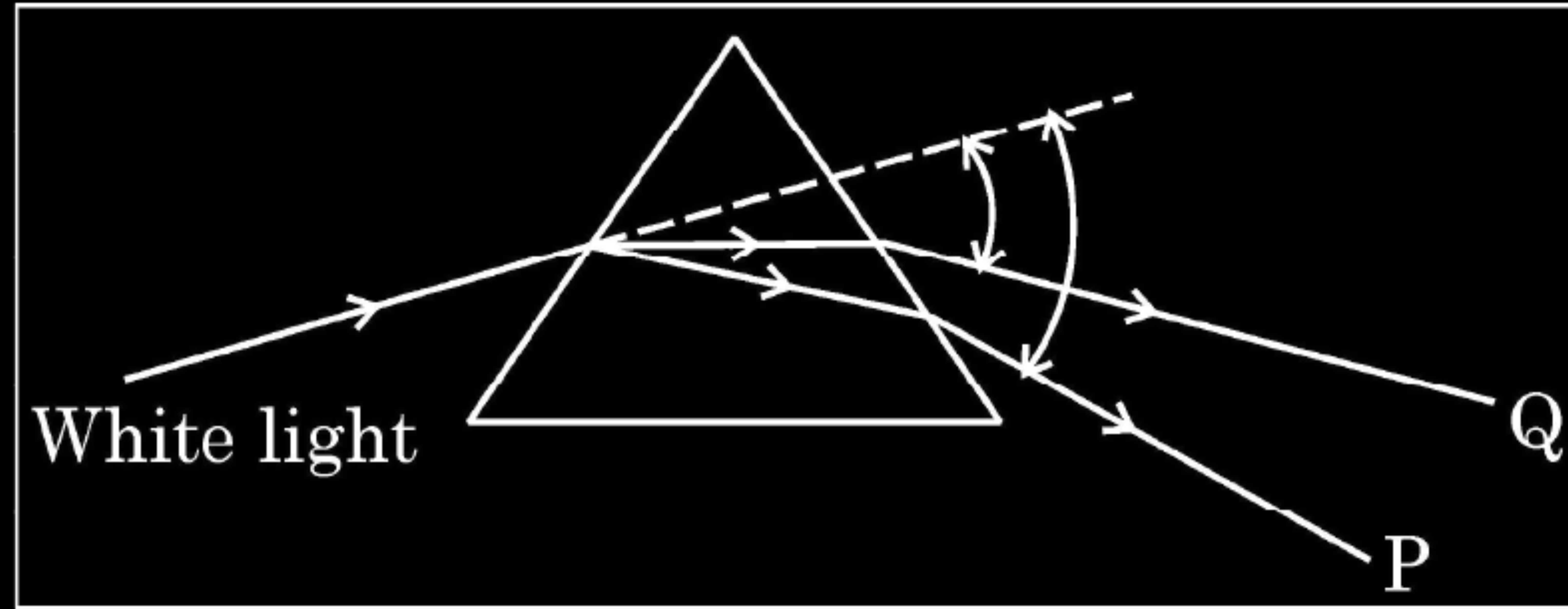
When sunlight passes through the atmosphere, fine particles in the air scatter blue light (shorter wavelengths) more than the red colour (longer wavelengths).

1



14. In the following diagram showing dispersion of white light by a glass prism, the colours 'P' and 'Q' respectively are –

1

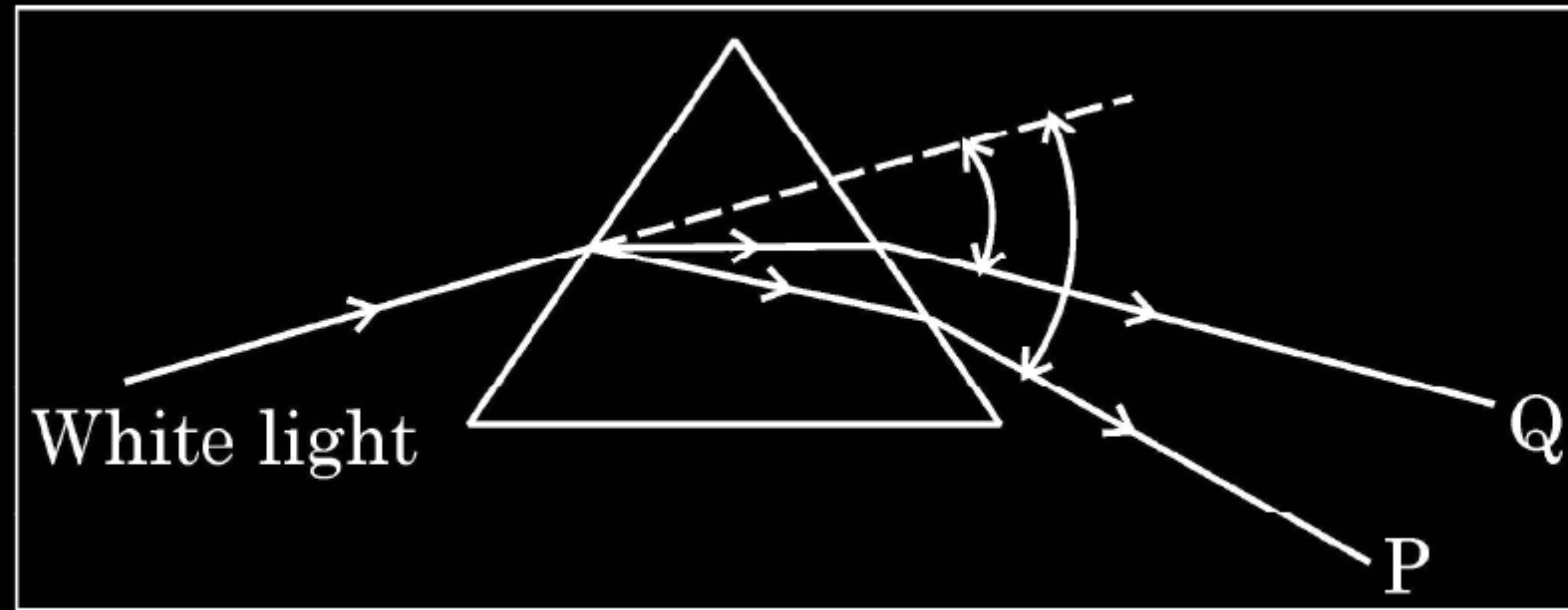


- |                    |                      |
|--------------------|----------------------|
| (a) Red and Violet | (b) Violet and Red   |
| (c) Red and Blue   | (d) Orange and Green |



14. In the following diagram showing dispersion of white light by a glass prism, the colours 'P' and 'Q' respectively are –

1



- |                    |                      |
|--------------------|----------------------|
| (a) Red and Violet | (b) Violet and Red   |
| (c) Red and Blue   | (d) Orange and Green |

(b) Violet and Red



13. The phenomena of light involved in the formation of rainbow are

1

- (a) Refraction, dispersion and scattering.
- (b) Refraction, reflection and dispersion.
- (c) Refraction, dispersion and internal reflection.
- (d) Reflection, dispersion and total internal reflection.



13. The phenomena of light involved in the formation of rainbow are

1

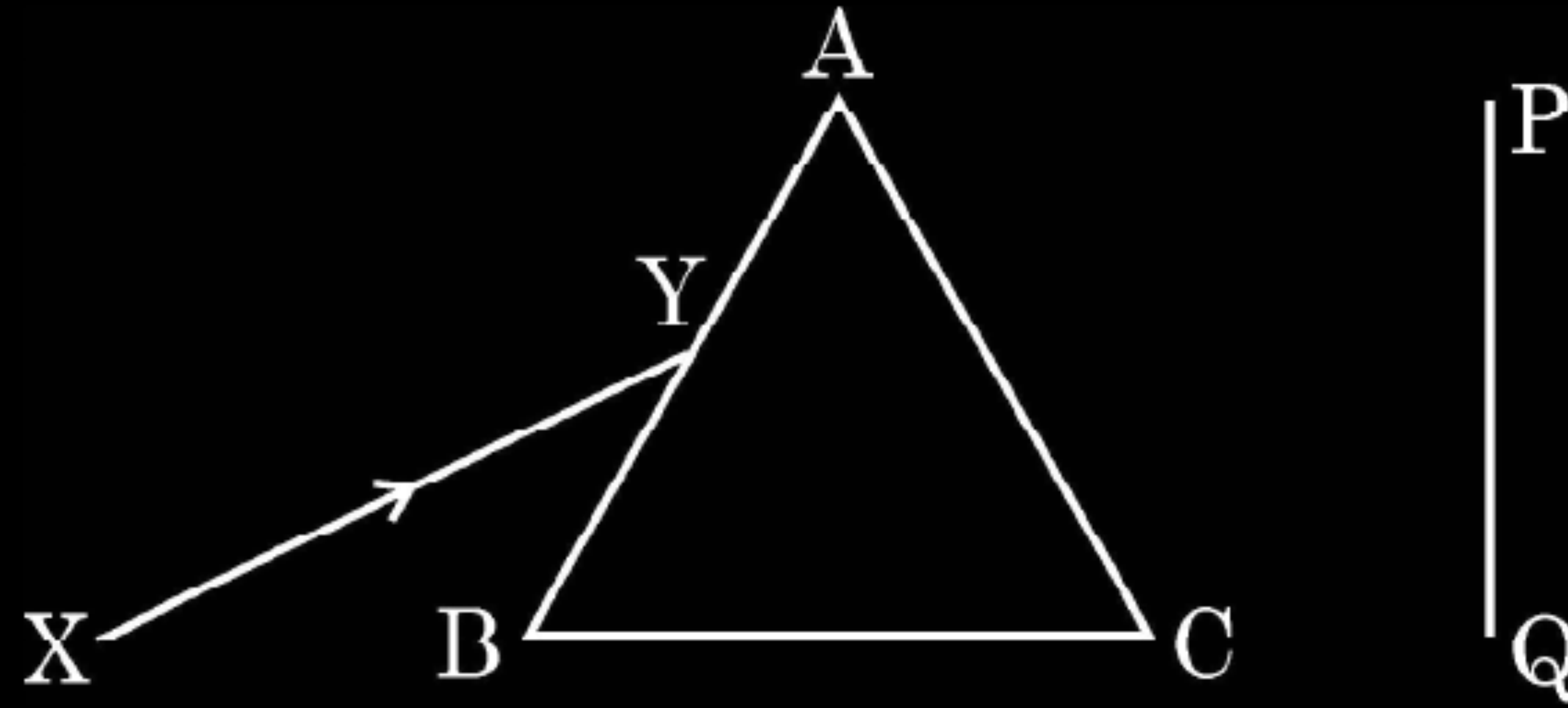
- (a) Refraction, dispersion and scattering.
- (b) Refraction, reflection and dispersion.
- (c) Refraction, dispersion and internal reflection.
- (d) Reflection, dispersion and total internal reflection.

(c) Refraction, dispersion and internal reflection.



26. (a) A narrow beam XY of white light is passing through a glass prism ABC as shown in the diagram :

2

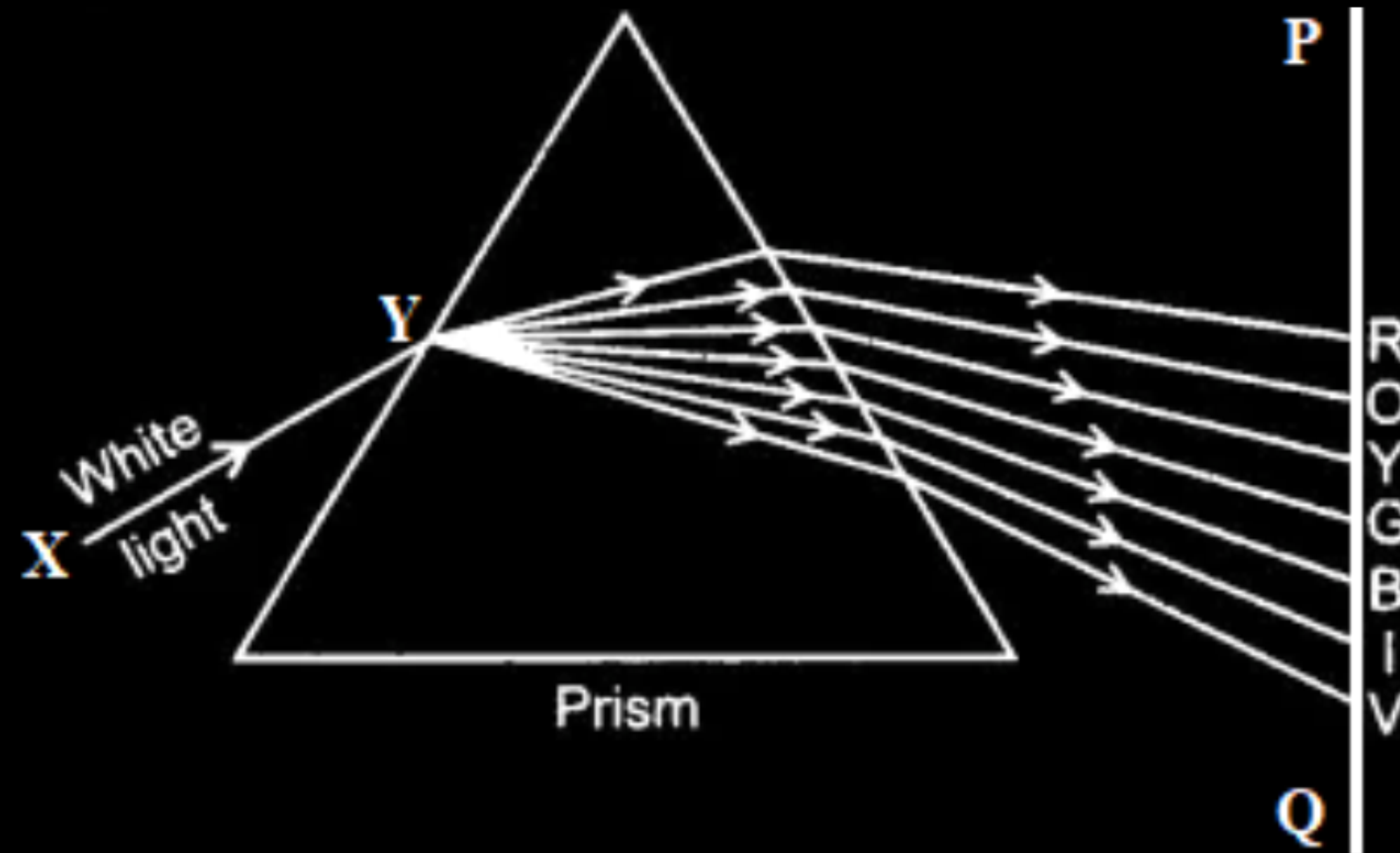


Trace it on your answer sheet and show the path of the emergent beam as observed on the screen PQ.

Name the phenomenon observed and state its cause.



(a)



- Dispersion of white light
- Cause: Different colours of light bend through different angles w.r.t. the incident ray. / Different colours have different wavelengths.

1

$\frac{1}{2}$

$\frac{1}{2}$



- 33. (A)** Define the term dispersion of white light. State the colour which bends (i) the most, (ii) the least while passing through a glass prism. Draw a diagram to show the dispersion of white light.

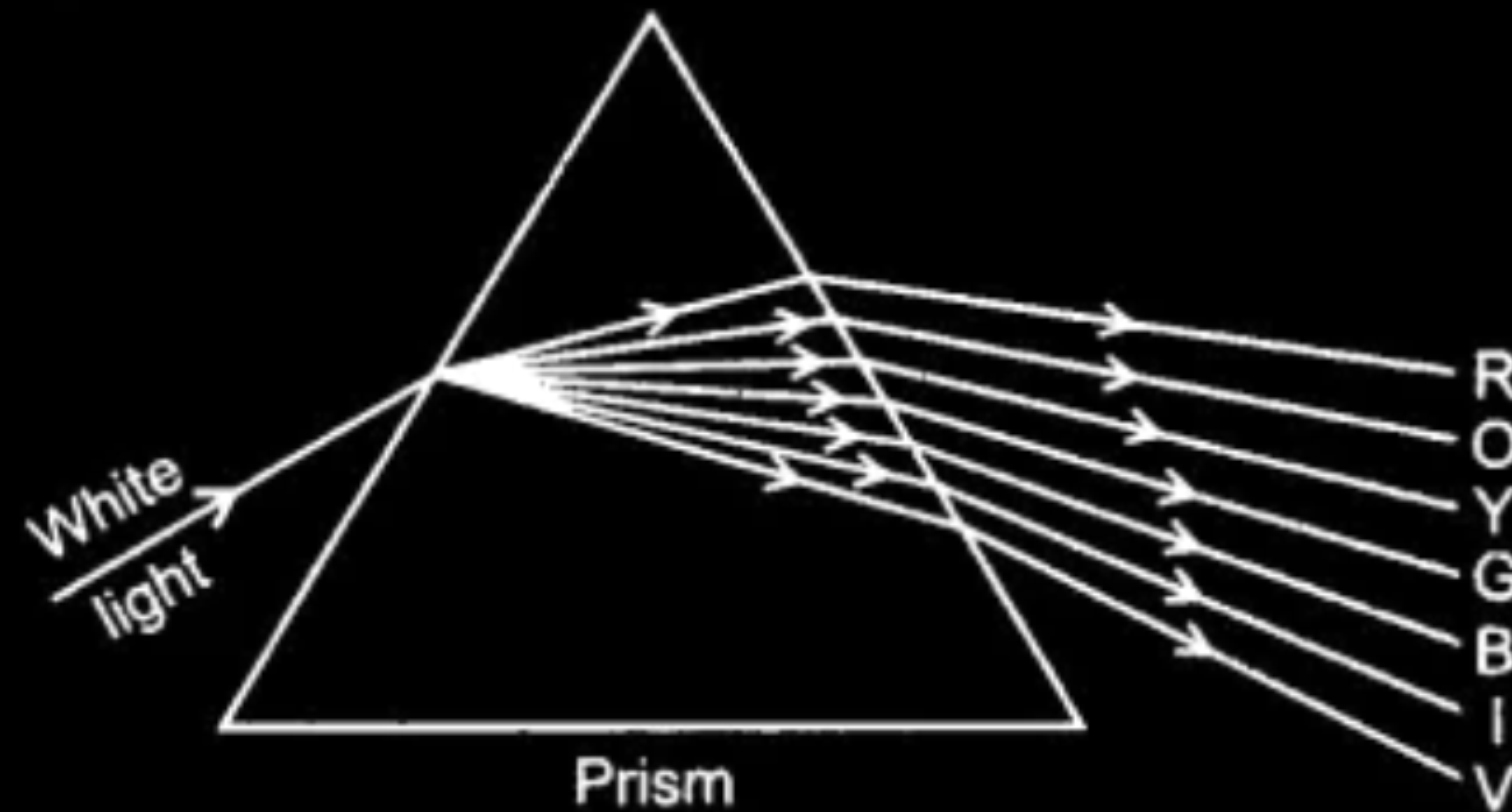


33. (A) Define the term dispersion of white light. State the colour which bends (i) the most, (ii) the least while passing through a glass prism. Draw a diagram to show the dispersion of white light.

3

(A)

- The splitting up of white light into its constituent/seven different colours while passing through a glass prism is called dispersion of white light.
- (i) Violet bends the most
- (ii) Red bends the least



1

$\frac{1}{2}$

$\frac{1}{2}$

1



**(B)** What is a rainbow ? Draw a labelled diagram to show its formation. **3**



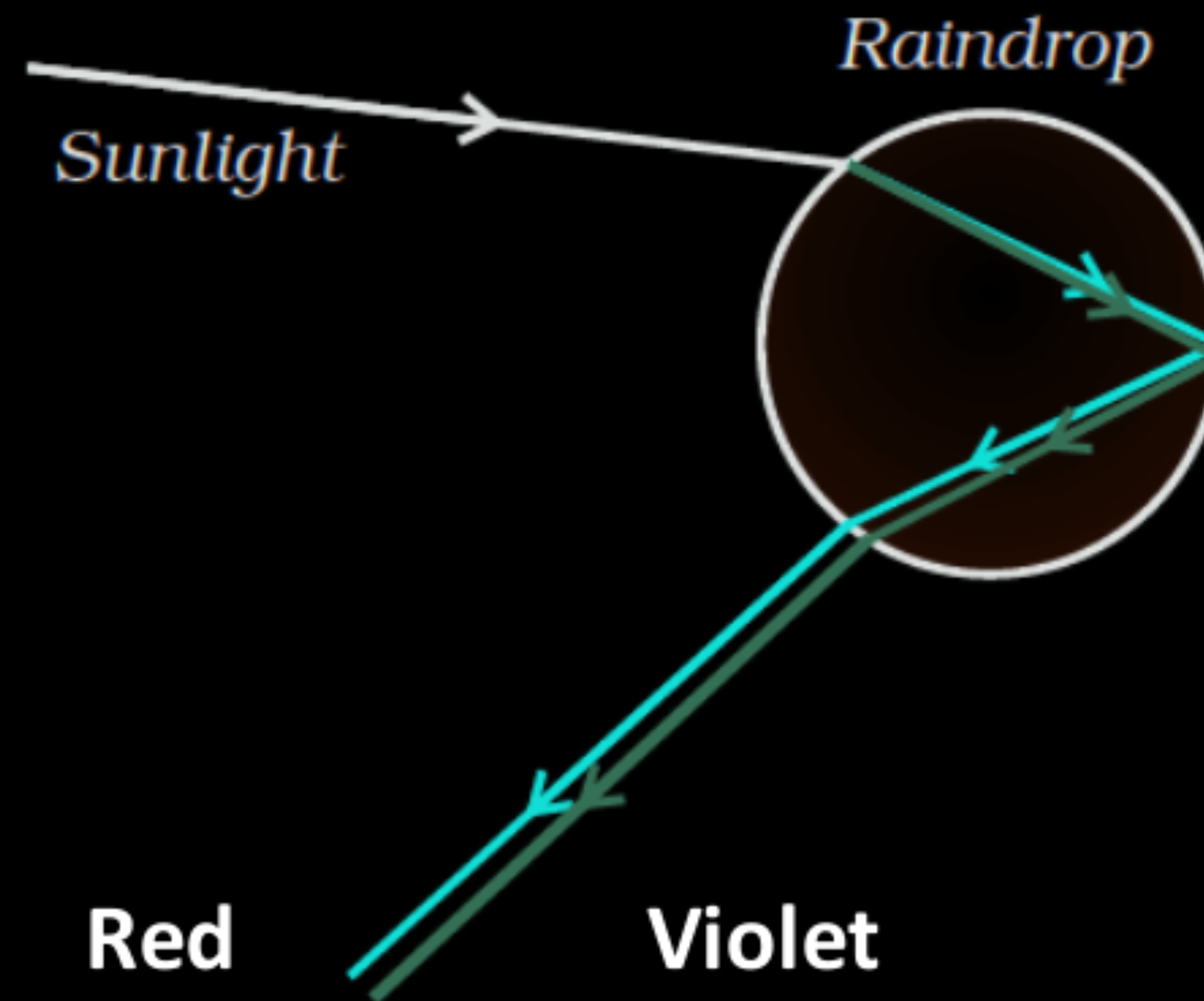
**(B)** What is a rainbow ? Draw a labelled diagram to show its formation.

3

**(B)**

- A rainbow is a natural spectrum of sunlight appearing in the sky after the rainfall.

- 



1

2



(b) What is dispersion of white light ? State its cause. Draw a diagram to show dispersion of a beam of white light by a glass prism.

3



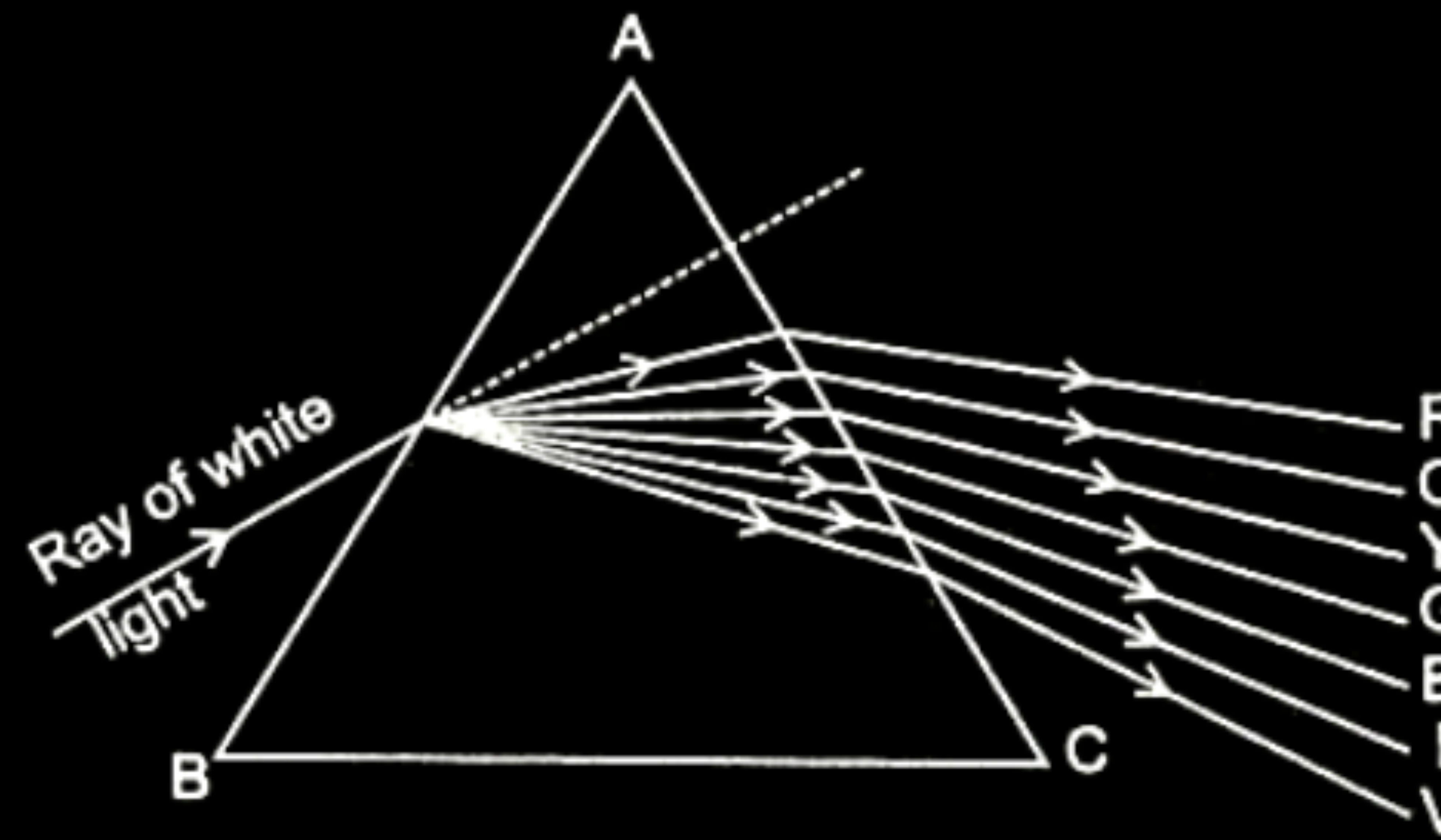
- (b) What is dispersion of white light ? State its cause. Draw a diagram to show dispersion of a beam of white light by a glass prism.

3

(b)

The splitting of white light into its constituent colours is called dispersion.

Cause: Different colours of white light bend through different angles with respect to incident ray.



1



**15.** The colour of light for which the refractive index of glass is minimum, is :

(A) Red

(B) Yellow

(C) Green

(D) Violet



**15.** The colour of light for which the refractive index of glass is minimum, is :

(A) Red

(B) Yellow

(C) Green

(D) Violet

(A) / Red



14. The phenomena of light involved in the formation of rainbow are :
- (A) Refraction, reflection and dispersion
  - (B) Refraction, dispersion and internal reflection
  - (C) Reflection, dispersion and internal reflection
  - (D) Refraction, dispersion, scattering and total internal reflection



14. The phenomena of light involved in the formation of rainbow are :
- (A) Refraction, reflection and dispersion
  - (B) Refraction, dispersion and internal reflection
  - (C) Reflection, dispersion and internal reflection
  - (D) Refraction, dispersion, scattering and total internal reflection

---

(B) / Refraction, Dispersion and internal reflection

---



**16.** When a narrow beam of white light passes through a glass prism it splits into its component colours ? This phenomenon is called :

- (a) Diffusion of light
- (b) Total reflection of light
- (c) Scattering of light
- (d) Dispersion of light



**16.** When a narrow beam of white light passes through a glass prism it splits into its component colours ? This phenomenon is called :

- (a) Diffusion of light
- (b) Total reflection of light
- (c) Scattering of light
- (d) Dispersion of light

(d) Dispersion of light



**17.**    *Assertion (A) :* A rainbow is an artificial spectrum of white light appearing in the sky after a rain shower.

*Reason (R) :*    The water droplets act like small prisms.



**17.**    *Assertion (A) :* A rainbow is an artificial spectrum of white light appearing in the sky after a rain shower.

*Reason (R) :*    The water droplets act like small prisms.

(d)    Assertion (A) is false, but Reason (R) is true.



- 31.** (a) White light is dispersed into seven visible coloured components by a glass prism. Name the colour which bends (i) the most and (ii) the least.
- (b) How can the coloured components of white light be recombined after a prism has separated them ? Draw a labelled diagram to justify your answer.

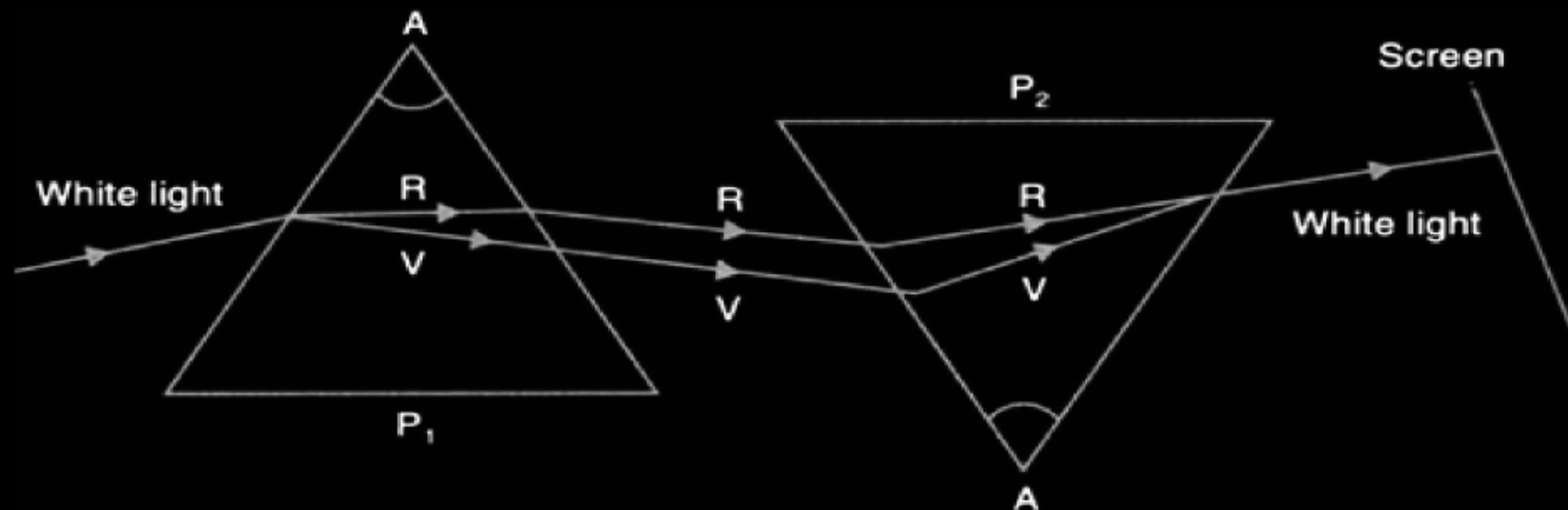


- 31.** (a) White light is dispersed into seven visible coloured components by a glass prism. Name the colour which bends (i) the most and (ii) the least.
- (b) How can the coloured components of white light be recombined after a prism has separated them ? Draw a labelled diagram to justify your answer.

3

- (a) (i) Violet  
(ii) Red

(b) By placing another identical prism in an inverted position with respect to the first prism.



$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$1\frac{1}{2}$



**20. Assertion (A) :** The rainbow is a natural spectrum of sunlight in the sky.

**Reason (R) :** Rainbow is formed in the sky when the sun is overhead and water droplets are also present in air.



**20. Assertion (A) :** The rainbow is a natural spectrum of sunlight in the sky.

**Reason (R) :** Rainbow is formed in the sky when the sun is overhead and water droplets are also present in air.

**1**

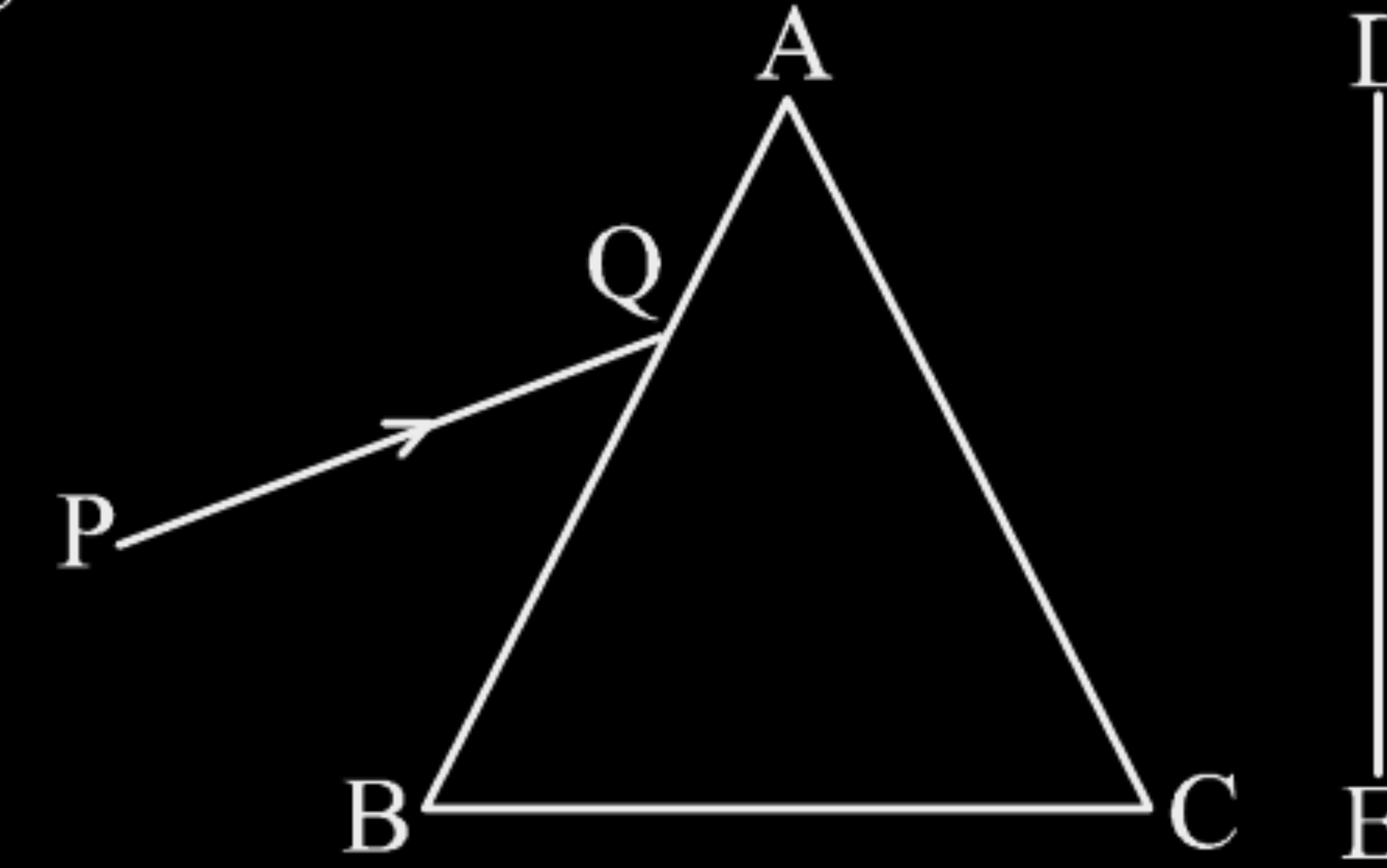
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(c) /Assertion (A) is true, but Reason (R) is false.

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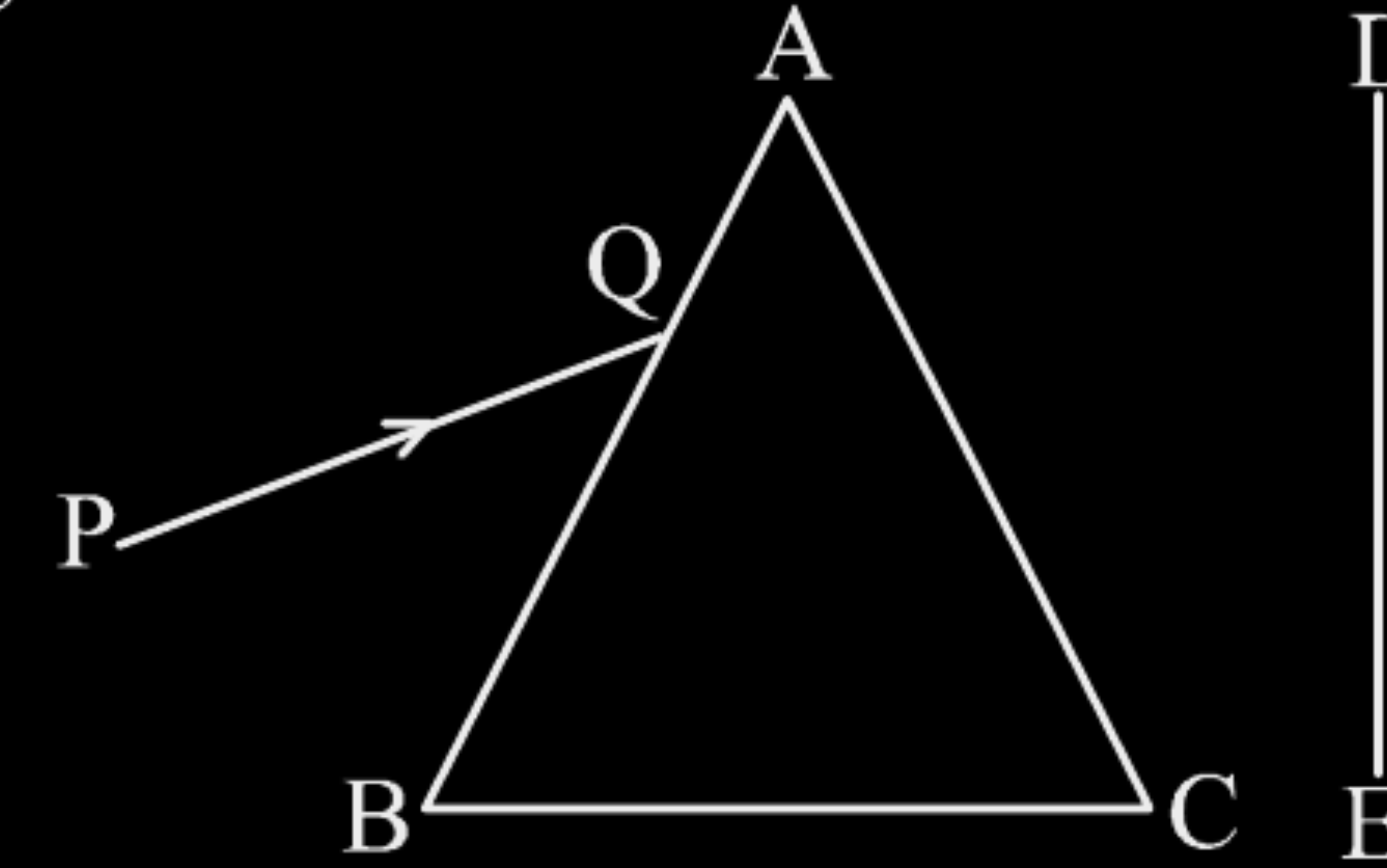
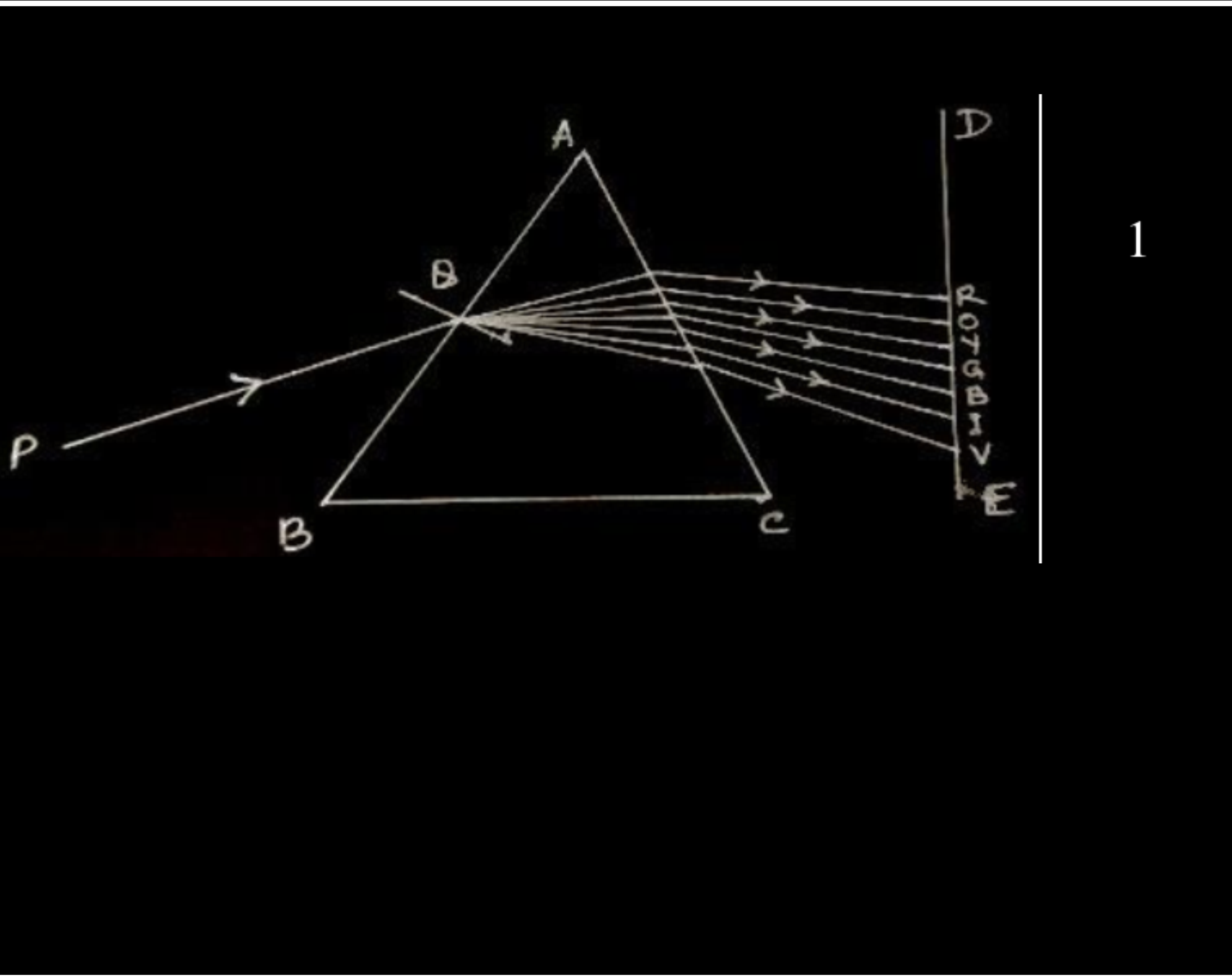
32. A narrow beam, PQ of white light is passing through a glass prism ABC as shown in the diagram.



Draw a ray diagram to show the emergent beam as it falls on the screen DE. Also write the phenomenon involved and its cause. Using the second law of refraction state which colour of light must have the highest value of refractive index amongst seven visible colours of light. Justify your answer.



32. A narrow beam, PQ of white light is passing through a glass prism ABC as shown in the diagram.



Draw a ray diagram to show the emergent beam as it falls on the screen DE. Also write the phenomenon involved and its cause. Using the second law of refraction state which colour of light must have the highest value of refractive index amongst seven visible colours of light. Justify your answer.

3

|                                                                                                                                                                                                        |               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| • Phenomenon: Dispersion of light                                                                                                                                                                      | $\frac{1}{2}$ |
| • Cause : Different colours of white light bend through different angles with respect to incident ray./ Different colours of white light have different wavelength therefore bend by different angles. | $\frac{1}{2}$ |
| • Refractive index of glass is highest for violet colour.                                                                                                                                              | $\frac{1}{2}$ |
| • Justification :For same $\angle i$ the $\angle r$ is minimum for the violet light.                                                                                                                   | $\frac{1}{2}$ |
| $\left(\frac{\sin i}{\sin r}\right)$ or refractive index is highest                                                                                                                                    |               |



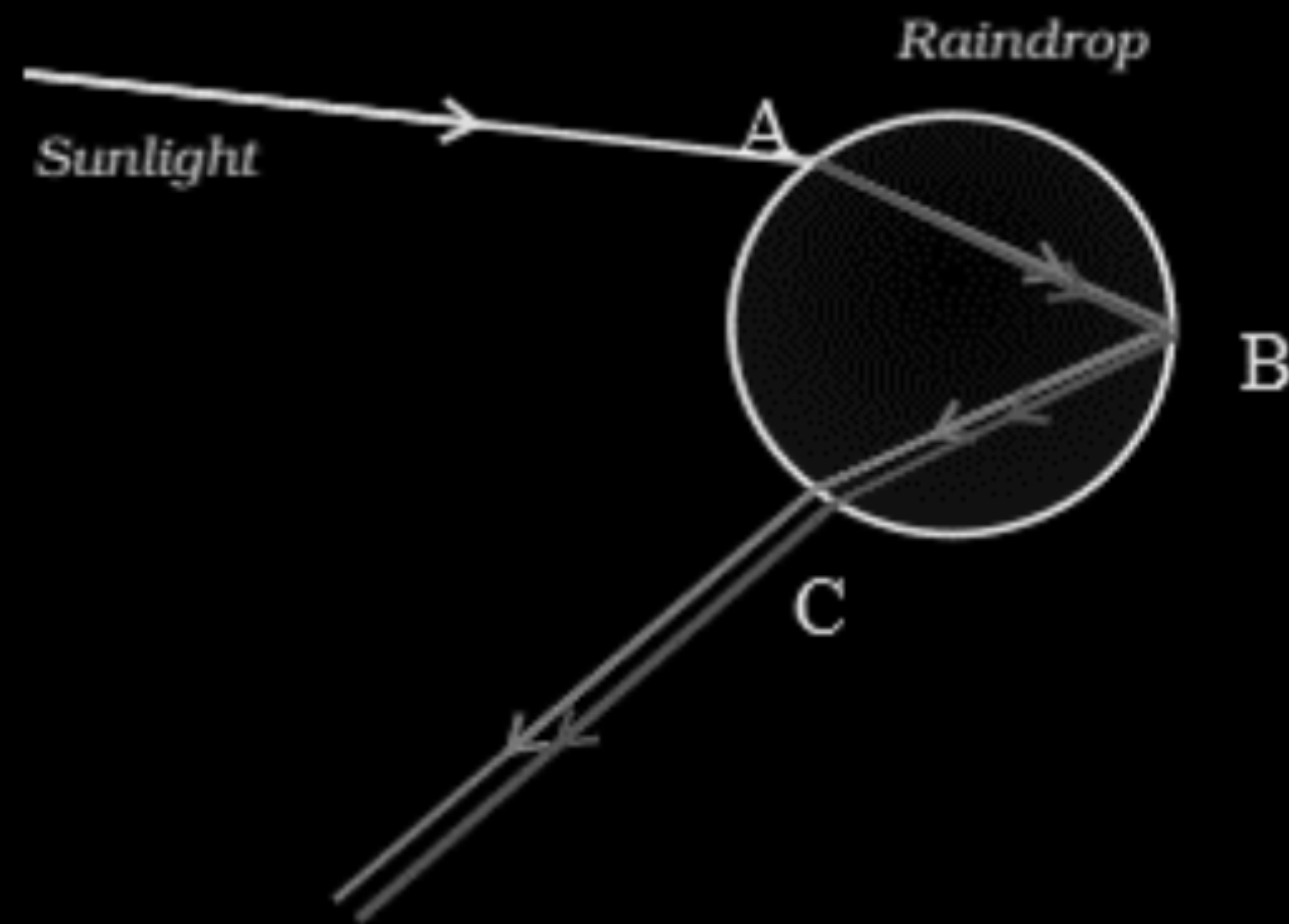
- (b) Draw a ray diagram to show the formation of a rainbow in the sky. On this diagram mark A – where dispersion of light occurs, B – where internal reflection of light occurs and C – where refraction of light occurs. List two necessary conditions to observe a rainbow.



- (b) Draw a ray diagram to show the formation of a rainbow in the sky. On this diagram mark A – where dispersion of light occurs, B – where internal reflection of light occurs and C – where refraction of light occurs. List two necessary conditions to observe a rainbow.

3

(b)



**3 Labellings (A, B, C)**

- Two conditions :
- (i) Presence of tiny water droplets in the atmosphere.
- (ii) Position of Sun at the back of/behind the observer.

$\frac{1}{2}$

$\frac{1}{2} \times 3$

$\frac{1}{2}$

$\frac{1}{2}$



13. The Phenomena of light involved in the formation of a rainbow in the sky are

1

- (A) Refraction, dispersion and reflection
- (B) Refraction, dispersion and total internal reflection
- (C) Dispersion, scattering and reflection
- (D) Dispersion, refraction and internal reflection



13. The Phenomena of light involved in the formation of a rainbow in the sky are

1

- (A) Refraction, dispersion and reflection
- (B) Refraction, dispersion and total internal reflection
- (C) Dispersion, scattering and reflection
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(D) Dispersion, refraction and internal reflection



**15.** The colour of light for which the refractive index of glass is minimum, is :

(A) Red

(B) Yellow

(C) Green

(D) Violet



**15.** The colour of light for which the refractive index of glass is minimum, is :

(A) Red

(B) Yellow

(C) Green

(D) Violet

---

(A) / Red

---



(b) What is dispersion of white light ? State its cause. Draw a diagram to show dispersion of a beam of white light by a glass prism.

3



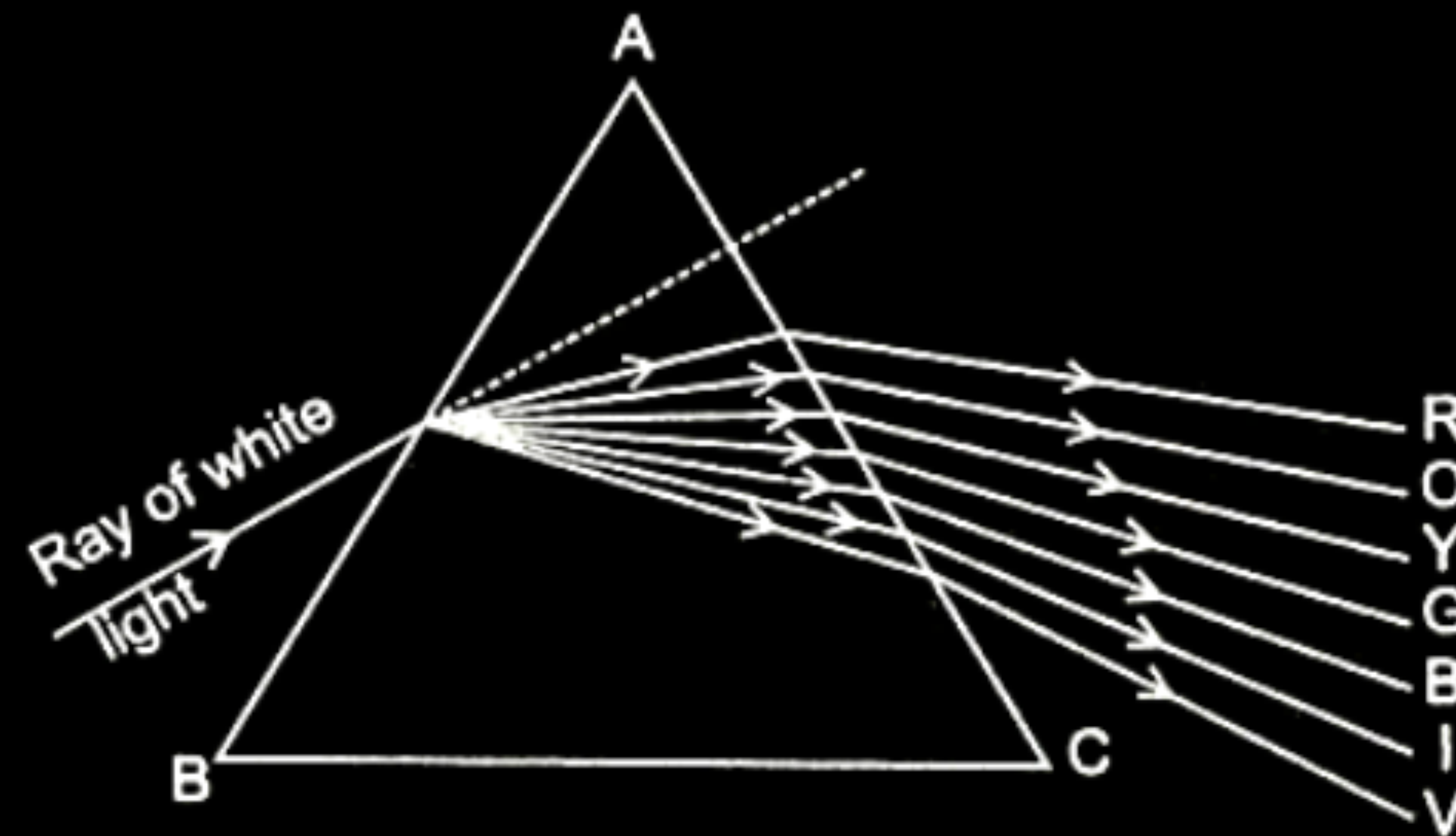
- (b) What is dispersion of white light ? State its cause. Draw a diagram to show dispersion of a beam of white light by a glass prism.

3

(b)

The splitting of white light into its constituent colours is called dispersion.

Cause: Different colours of white light bend through different angles with respect to incident ray.



1



**28.** When and where does a rainbow appear in the sky ? List the phenomena of light in proper sequence which occurs during the formation of a rainbow. Why can a rainbow not be seen from the surface of the moon ?



**28.** When and where does a rainbow appear in the sky ? List the phenomena of light in proper sequence which occurs during the formation of a rainbow. Why can a rainbow not be seen from the surface of the moon ?

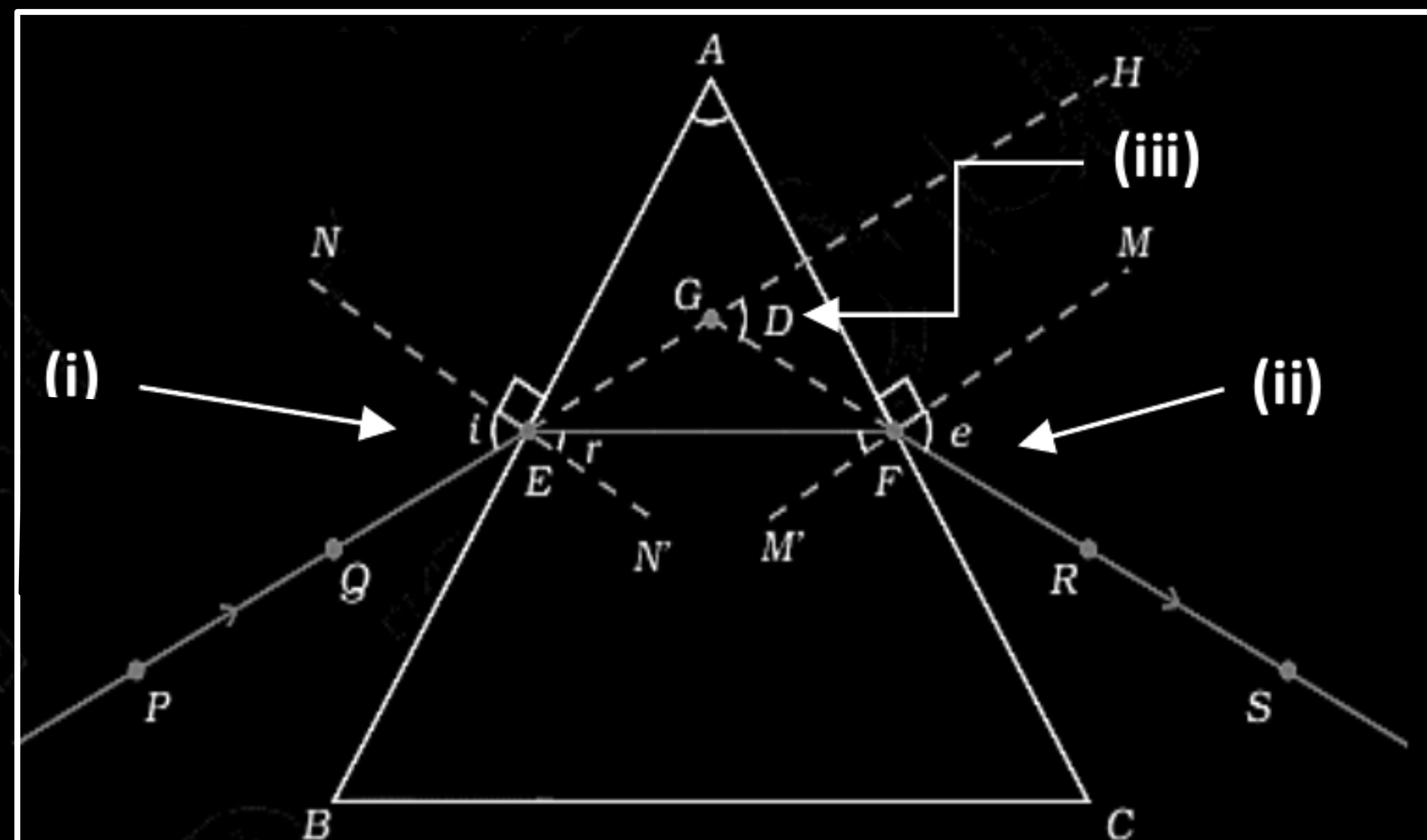
3

|   |                                                                                       |                        |
|---|---------------------------------------------------------------------------------------|------------------------|
| • | After the rain shower during day time                                                 | $\frac{1}{2}$          |
| • | In the sky opposite to the Sun                                                        | $\frac{1}{2}$          |
| • | Refraction; dispersion of light; Internal reflection                                  | $\frac{1}{2} \times 3$ |
| • | Because there is no atmosphere (water droplets) for the refraction of light to occur. | $\frac{1}{2}$          |



33. Draw a ray diagram to show the refraction of a ray of light through a triangular glass prism and mark on it (i) angle of incidence ( $i$ ), (ii) angle of emergence ( $e$ ), and (iii) angle of deviation ( $D$ ). What would happen, if instead of a ray, a beam of white light falls on the prism ? Give reason for the phenomenon that occurs in this case.

3



[Fig 10.4, page 166 of NCERT]

Labelled diagram

1 ½

- White light will split into a band of seven colours of spectrum.
- Different colours of white light bend / refract through different angles with respect to incident ray as they pass through a prism.

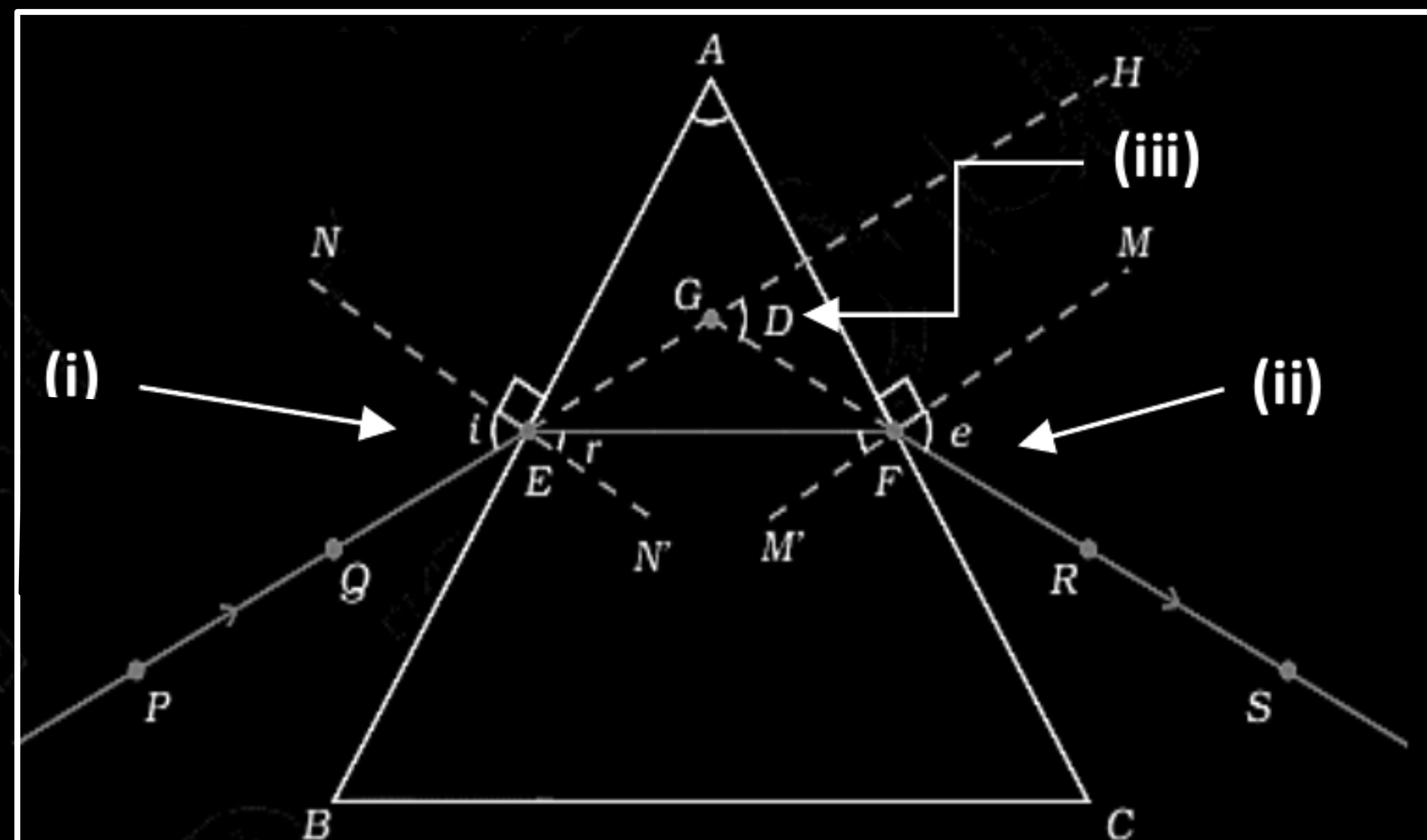
½

1



33. Draw a ray diagram to show the refraction of a ray of light through a triangular glass prism and mark on it (i) angle of incidence ( $i$ ), (ii) angle of emergence ( $e$ ), and (iii) angle of deviation ( $D$ ). What would happen, if instead of a ray, a beam of white light falls on the prism ? Give reason for the phenomenon that occurs in this case.

3



[Fig 10.4, page 166 of NCERT]

Labelled diagram

1 ½

- White light will split into a band of seven colours of spectrum.
- Different colours of white light bend / refract through different angles with respect to incident ray as they pass through a prism.

½

1



- 23.** (a) Draw a ray diagram to show the formation of a rainbow in the sky. Mark on the diagram three points A, B and C as given below :

2

A – where dispersion of light occurs

B – where internal reflection of dispersed light occurs

C – where refraction of dispersed light occurs



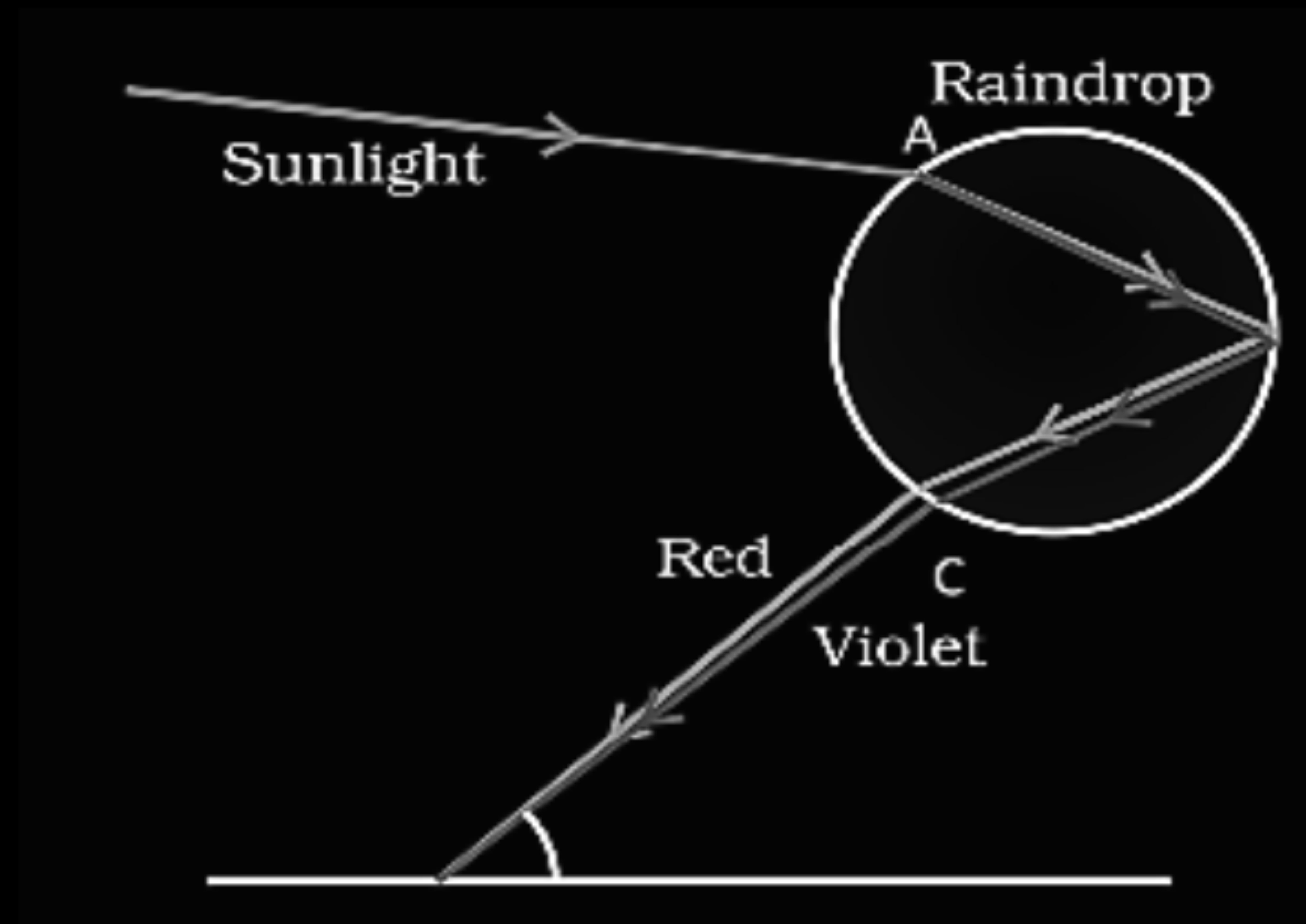
23. (a) Draw a ray diagram to show the formation of a rainbow in the sky. Mark on the diagram three points A, B and C as given below : 2

A – where dispersion of light occurs

B – where internal reflection of dispersed light occurs

C – where refraction of dispersed light occurs

(a)



[Fig 10.8 page 16]

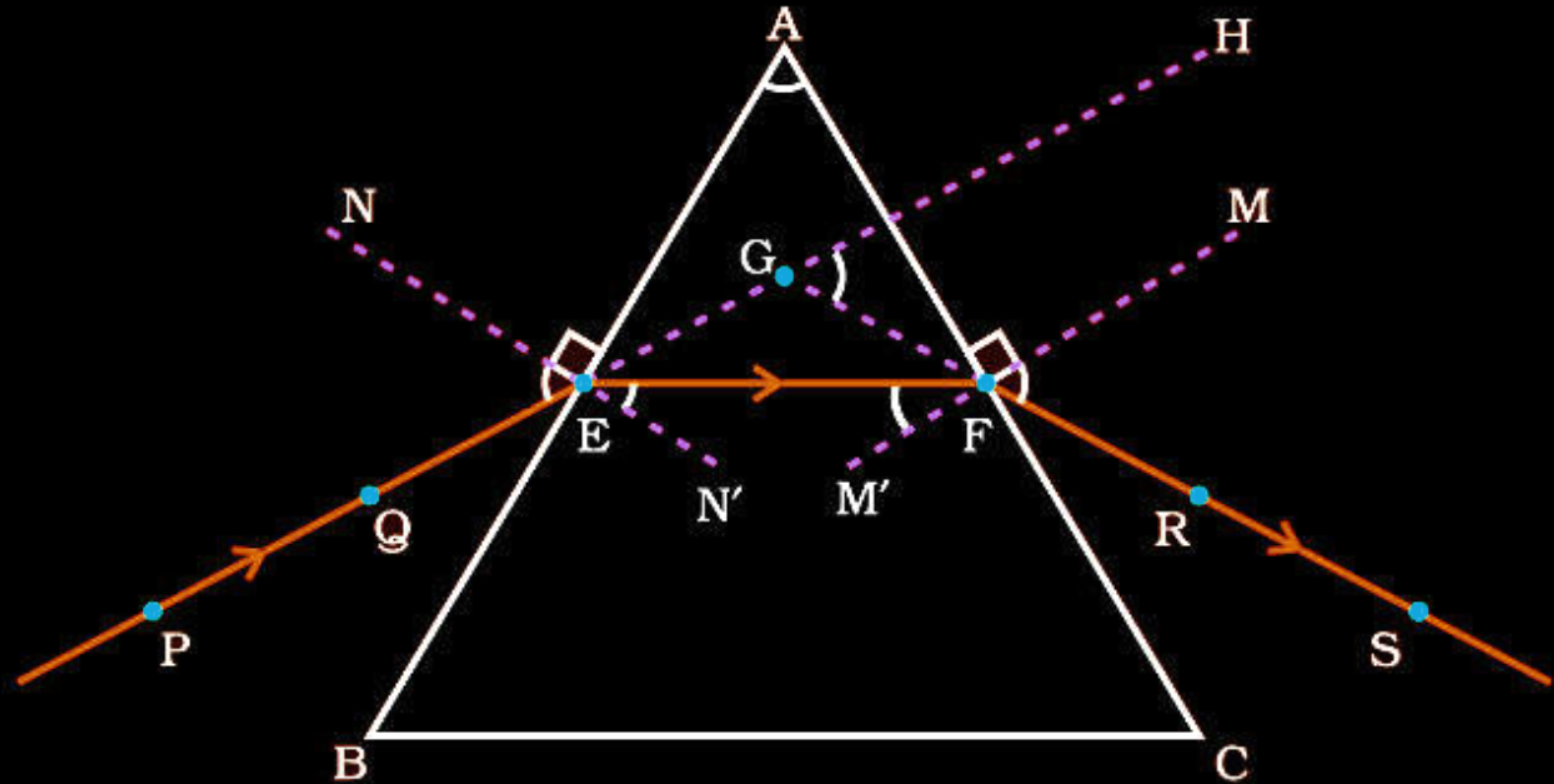
Diagram 1/2

Correct labelling of A,B and C 1 1/2



33. (a) Study the following ray diagram and name the angles given below :

$\angle PEN$ ,  $\angle FEN'$ ,  $\angle SFM$  and  $\angle FGH$



- (b) In this diagram, if  $\frac{\sin \angle PEN}{\sin \angle FEN'} = \frac{3}{2}$ , what is the value of  $\frac{\sin \angle SFM}{\sin \angle EFM}$  ?

Give reason to justify your answer.



(a)

$\angle PEN = \angle i$  / angle of incidence

$\frac{1}{2}$

$\angle FEN' = \angle r$  / angle of refraction

$\frac{1}{2}$

$\angle SFM = \angle e$  / angle of emergence

$\frac{1}{2}$

$\angle FGH = \angle D$  / angle of deviation

$\frac{1}{2}$

(b)

At face, AC of the prism

$$\frac{\sin i}{\sin r} = \frac{\sin \angle EFM'}{\sin \angle SFM} = \frac{2}{3}$$

$\therefore$

$$\frac{\sin \angle SFM}{\sin \angle EFM'} = \frac{3}{2}$$

Reason: Path of light is reversed at the face AC of the prism.

1

**Note:- Give one mark for part (b) to all students as in question**

**given,  $\frac{\sin \angle SFM}{\sin \angle EFM}$  is wrong. It should be  $\frac{\sin \angle SFM}{\sin \angle EFM'}$**



32. (a) Define the term dispersion of white light. State the colour of light which bends (i) the least, (ii) the most, while passing through a glass prism. Also, state the cause of dispersion of a narrow beam of white light when it passes through a prism.



32. (a) Define the term dispersion of white light. State the colour of light which bends (i) the least, (ii) the most, while passing through a glass prism. Also, state the cause of dispersion of a narrow beam of white light when it passes through a prism.

3

|                                                                                                                                                                                           |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <ul style="list-style-type: none"> <li>• Splitting of white light into its component colours when it passes through a prism/rain drop/oil bubble.</li> </ul>                              | 1             |
| <ul style="list-style-type: none"> <li>• Red colour bends the least</li> </ul>                                                                                                            | $\frac{1}{2}$ |
| <ul style="list-style-type: none"> <li>• Violet colour bends the most.</li> </ul>                                                                                                         | $\frac{1}{2}$ |
| <ul style="list-style-type: none"> <li>• Different colours of white light bend/refract through different angles with respect to the incident ray as they pass through a prism.</li> </ul> | 1             |



- (b) When and where does a rainbow appear in the sky ? Write sequence-wise the phenomena of light responsible for the formation of a rainbow. List two essential conditions for observing a rainbow.



(b) When and where does a rainbow appear in the sky ? Write sequence-wise the phenomena of light responsible for the formation of a rainbow. List two essential conditions for observing a rainbow.

3

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <p>(b )</p> <ul style="list-style-type: none"> <li>•             <ul style="list-style-type: none"> <li>➤ After the rain shower.</li> <li>➤ In a direction opposite to that of the sun.</li> </ul> </li> <li>• The tiny water droplets             <ul style="list-style-type: none"> <li>➤ refract and disperse sunlight then</li> <li>➤ reflect it internally and finally</li> <li>➤ refract it again</li> </ul> <p>when it comes out of the rain drop.</p> </li> <li>• Two necessary condition –             <ul style="list-style-type: none"> <li>(i) Presence of tiny rain droplets in air.</li> <li>(ii) Sun behind the observer.</li> </ul> </li> </ul> | <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p>1</p> <p><math>\frac{1}{2} + \frac{1}{2}</math></p> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|



**37.** The students in a class took a thick sheet of cardboard and made a small hole in its centre. Sunlight was allowed to fall on this small hole and they obtained a narrow beam of white light. A glass prism was taken and this white light was allowed to fall on one of its faces. The prism was turned slowly until the light that comes out of the opposite face of the prism appeared on the nearby screen. They studied this beautiful band of light and concluded that it is a spectrum of white light.

- (i) Give any one more instance in which this type of spectrum is observed. **1**
- (ii) What happens to white light in the above case? **1**
- (iii) (A) List two conditions necessary to observe a rainbow. **2**

**OR**

- (iii) (B) Draw a ray diagram to show the formation of a rainbow. Mark on it, points (a), (b) and (c) as given below : **2**
  - (a) Where dispersion of light occurs.
  - (b) Where light gets reflected internally.
  - (c) Where final refraction occurs.



(i) A rainbow (or any other)

(ii) Dispersion of white light takes place.

(iii) (A)

- The presence of water droplets in the atmosphere.
- The sun must be at the back of the observer.

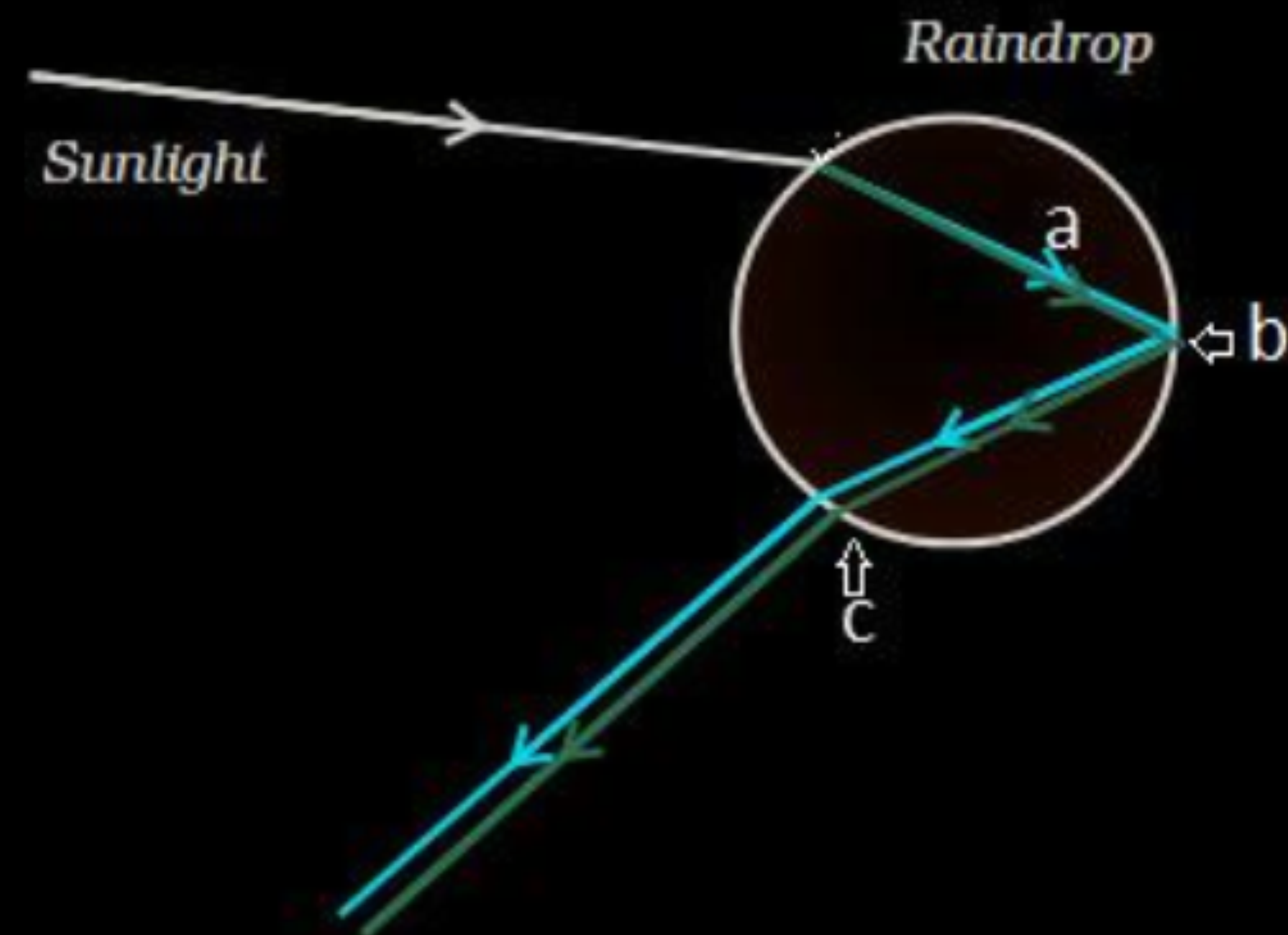
1

1

1+1

**OR**

(iii) (B)



$\frac{1}{2} \times 4$

( $\frac{1}{2}$  mark for diagram and  $\frac{1}{2}$  for labelling a, b, c)



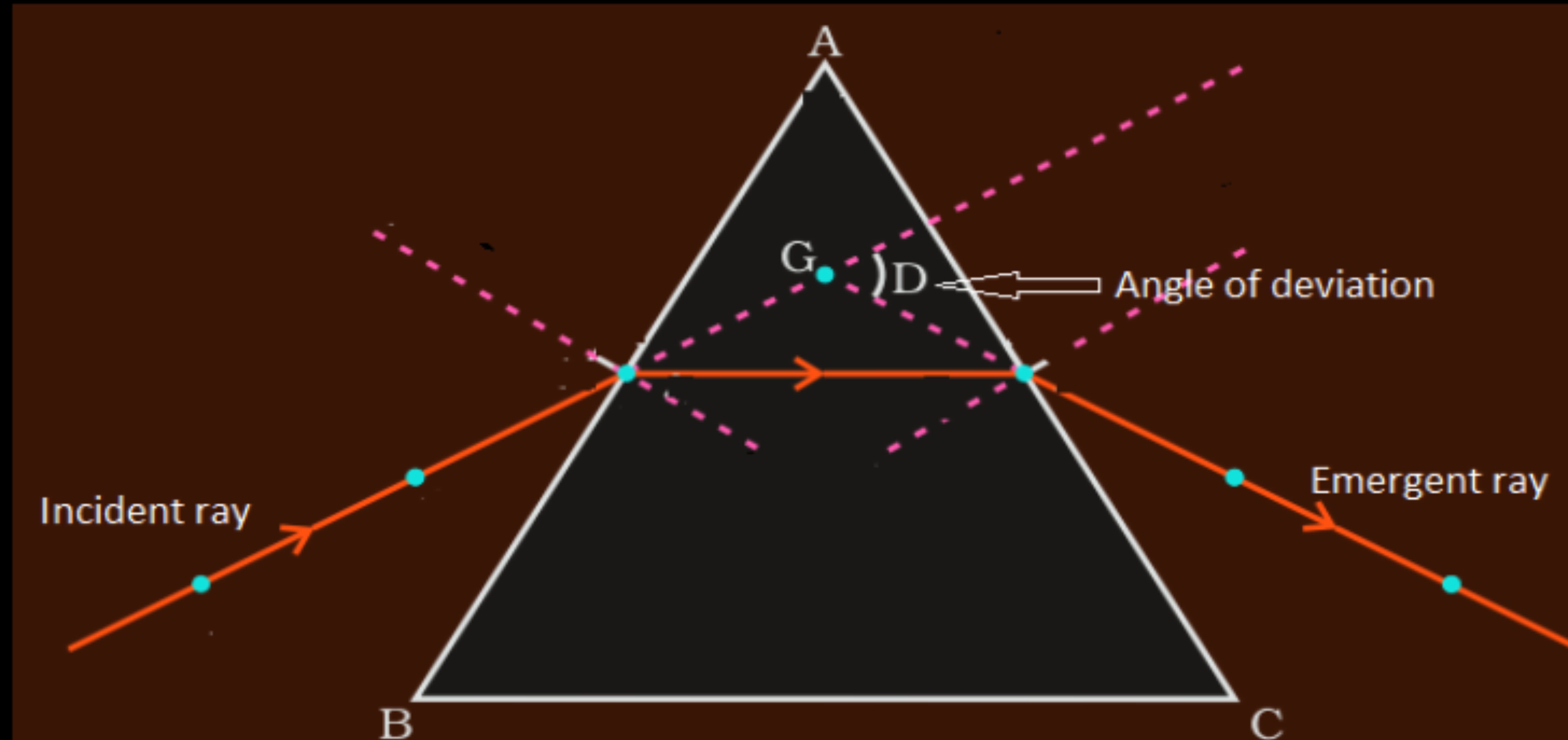
- 25. (A)** Draw a ray diagram to show the refraction of a ray of light passing through an equilateral glass prism. Mark the angle through which the emergent ray bends from the direction of the incident ray and also name it.



25. (A) Draw a ray diagram to show the refraction of a ray of light passing through an equilateral glass prism. Mark the angle through which the emergent ray bends from the direction of the incident ray and also name it.

2

(A)



(one mark for diagram and  $\frac{1}{2}$  for labelling.)

- Angle of deviation

$1\frac{1}{2}$

$\frac{1}{2}$



**20. Assertion (A) :** White light is dispersed by a glass prism into seven colours.

**Reason (R) :** The red light bends the least while the violet the most when a beam of white light passes through a glass prism.



**20. Assertion (A) :** White light is dispersed by a glass prism into seven colours.

**Reason (R) :** The red light bends the least while the violet the most when a beam of white light passes through a glass prism.

**1**

---

(b) / Both Assertion and Reason are true and Reason (R) is not the correct explanation of Assertion (A).

---



23. (a) Give reasons :

(i) The sky appears dark to passengers flying at very high altitude.

(ii) 'Danger' signal lights are red in colour.

**2**

**OR**

(b) What is a rainbow ? "We see a rainbow in the sky only after the rainfall." Why ?

**2**



23. (a) Give reasons :

- (i) The sky appears dark to passengers flying at very high altitude.
- (ii) 'Danger' signal lights are red in colour.

2

**OR**

(b) What is a rainbow ? "We see a rainbow in the sky only after the rainfall." Why ?

2

(a) (i) Scattering of light is not prominent at such heights.

1

(ii) The red colour is least scattered by smoke or fog. / Red colour has longer wavelength.

1

**OR**

(b)

- Rainbow is a natural spectrum appearing in the sky after a rain shower.

1

After rain, small water droplets act as a tiny prism, when light enters, it gets refracted and dispersed.

1



33. (a) Draw a diagram to show the dispersion of white light by a glass prism.

(b) What is spectrum ? Give reason for its formation.

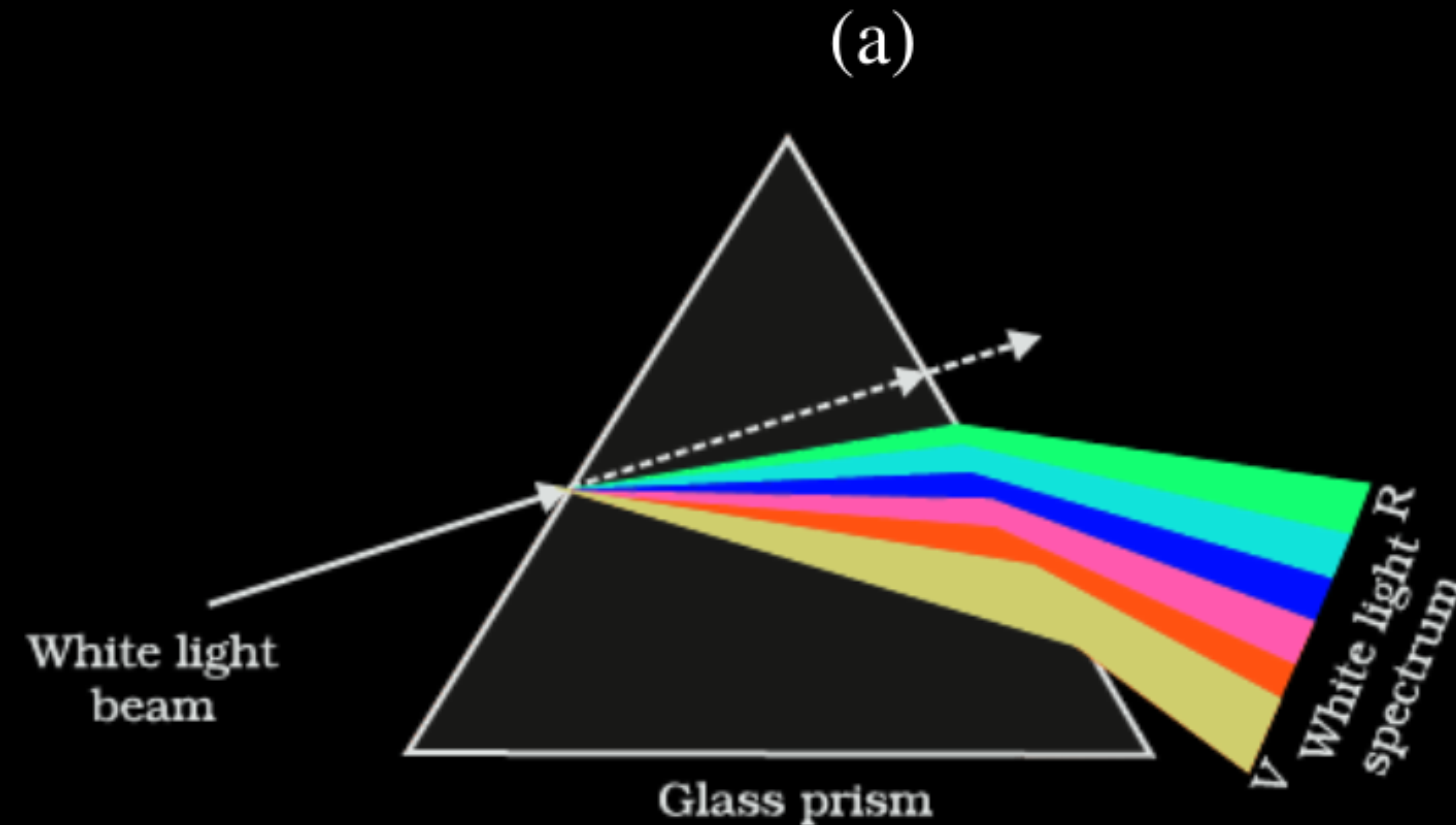
**3**



33. (a) Draw a diagram to show the dispersion of white light by a glass prism.

(b) What is spectrum ? Give reason for its formation.

3



1

(b) The band of seven colours obtained on the screen after dispersion of white light

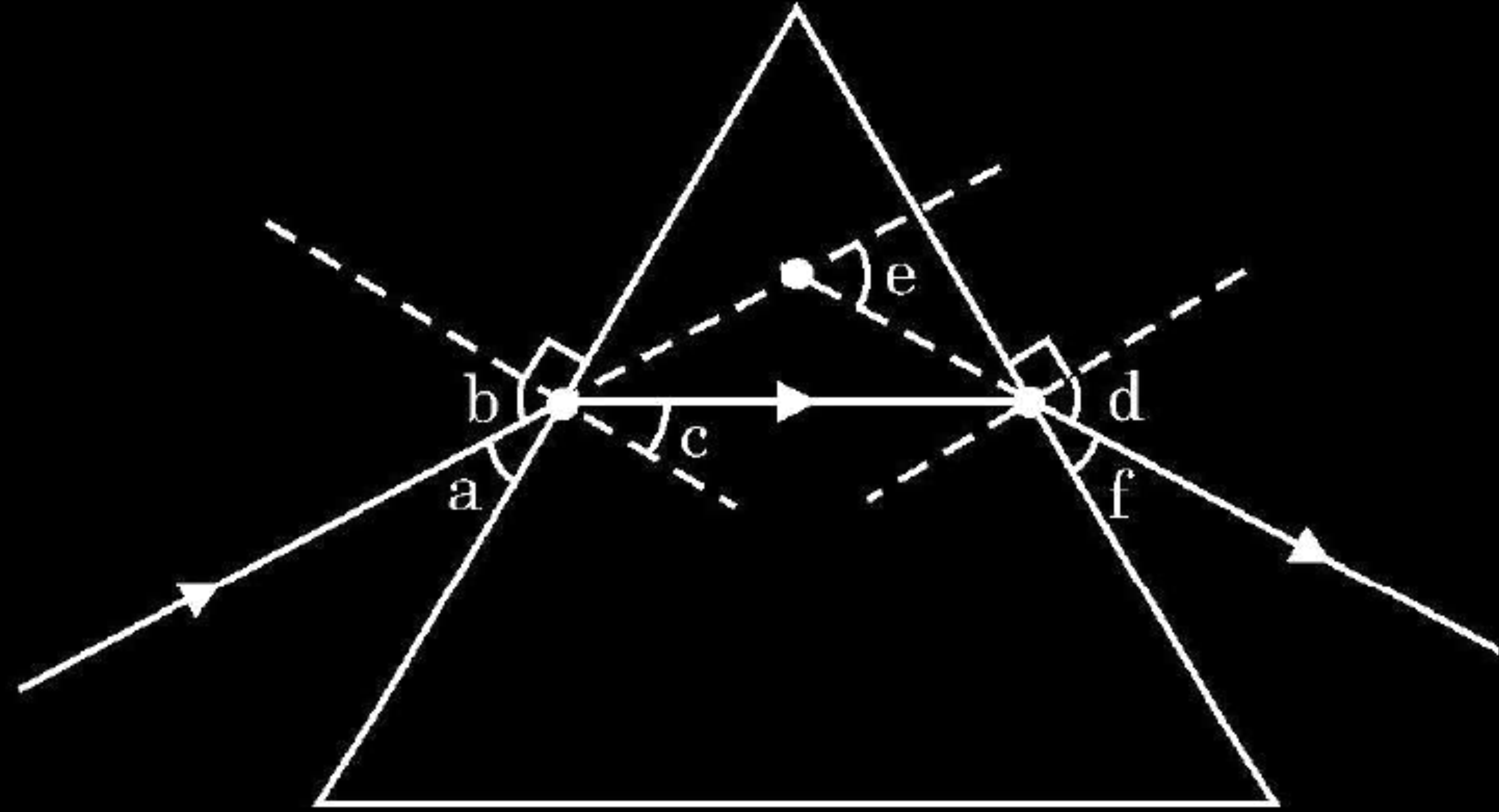
1

Reason: Different components of white light (colours) bend through different angles with respect to the incident ray as they pass through a prism

1



15. Consider the following ray diagram :



Here, the angle of incidence and angle of deviation respectively are

(A) a and e

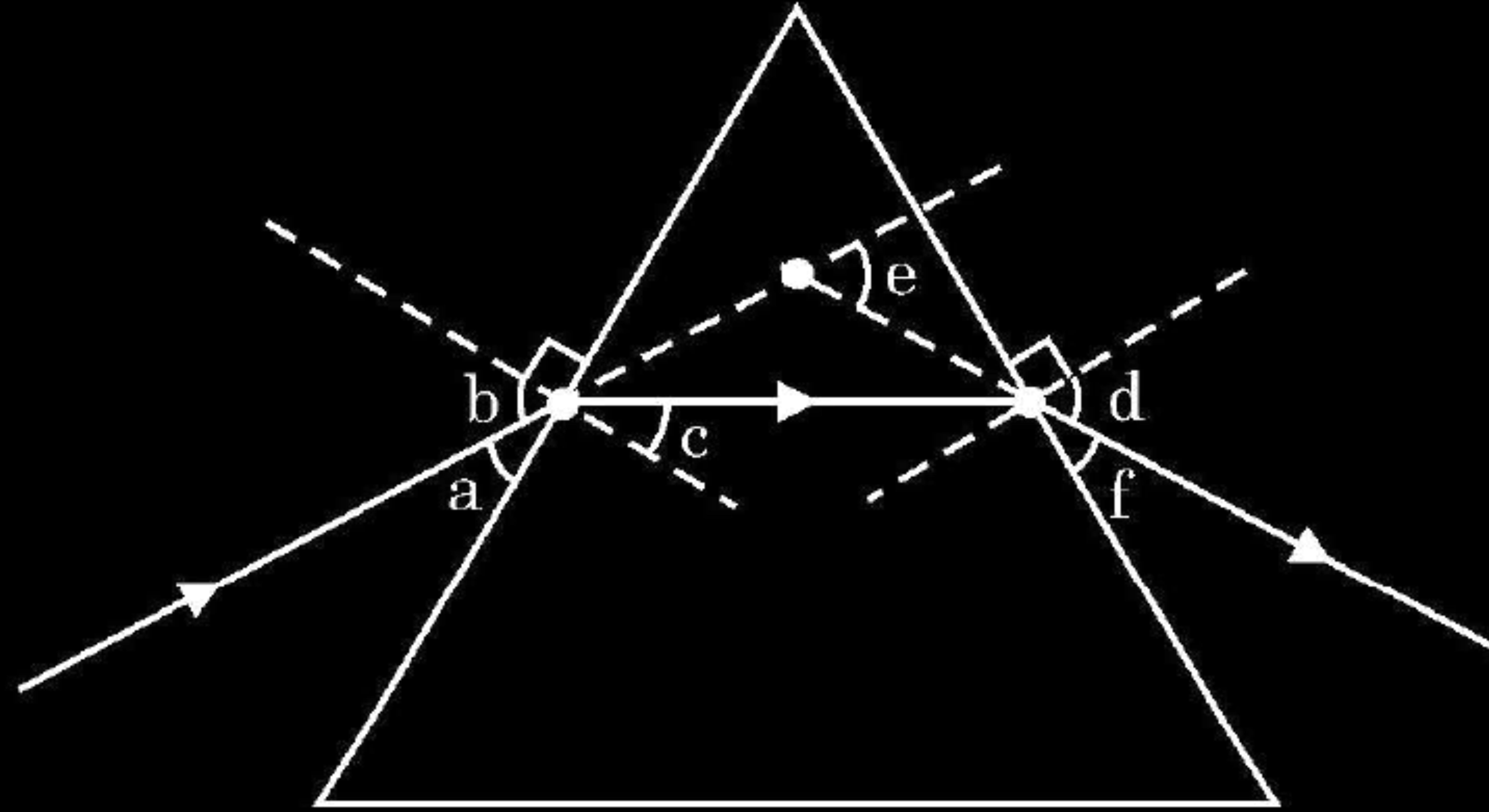
(B) b and d

(C) b and e

(D) a and f



15. Consider the following ray diagram :



Here, the angle of incidence and angle of deviation respectively are

(A) a and e

(B) b and d

(C) b and e

(D) a and f

(C) / b and e



- (b) Draw a ray diagram to show the path of a ray of light falling obliquely on one of the refracting faces of a triangular glass prism and mark the angle of deviation on it.



- (b) Draw a ray diagram to show the path of a ray of light falling obliquely on one of the refracting faces of a triangular glass prism and mark the angle of deviation on it.

2

(b)

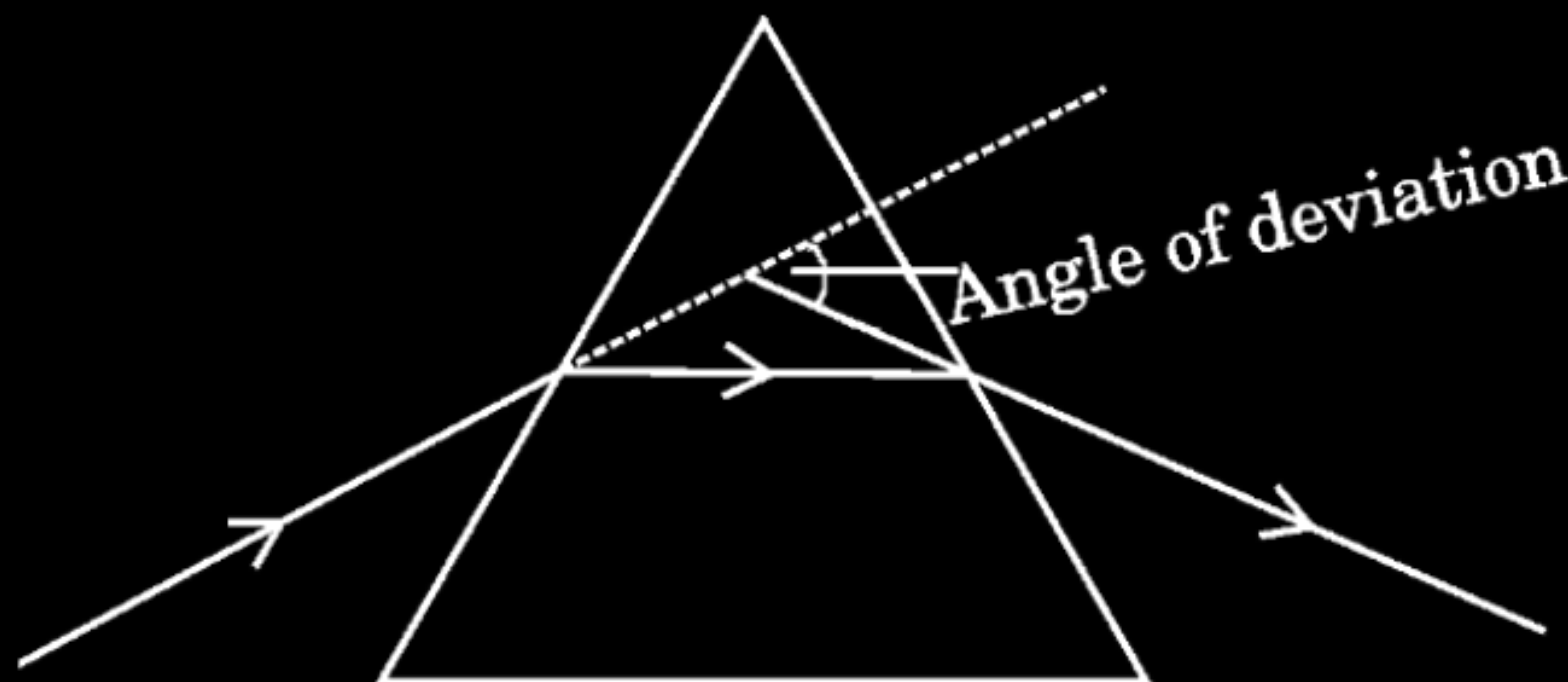


DIAGRAM  
DIRECTION OF RAYS  
MARKING OF ANGLE

1

$\frac{1}{2}$

$\frac{1}{2}$



**31.** What is a rainbow ? Draw a labelled diagram to show its formation.

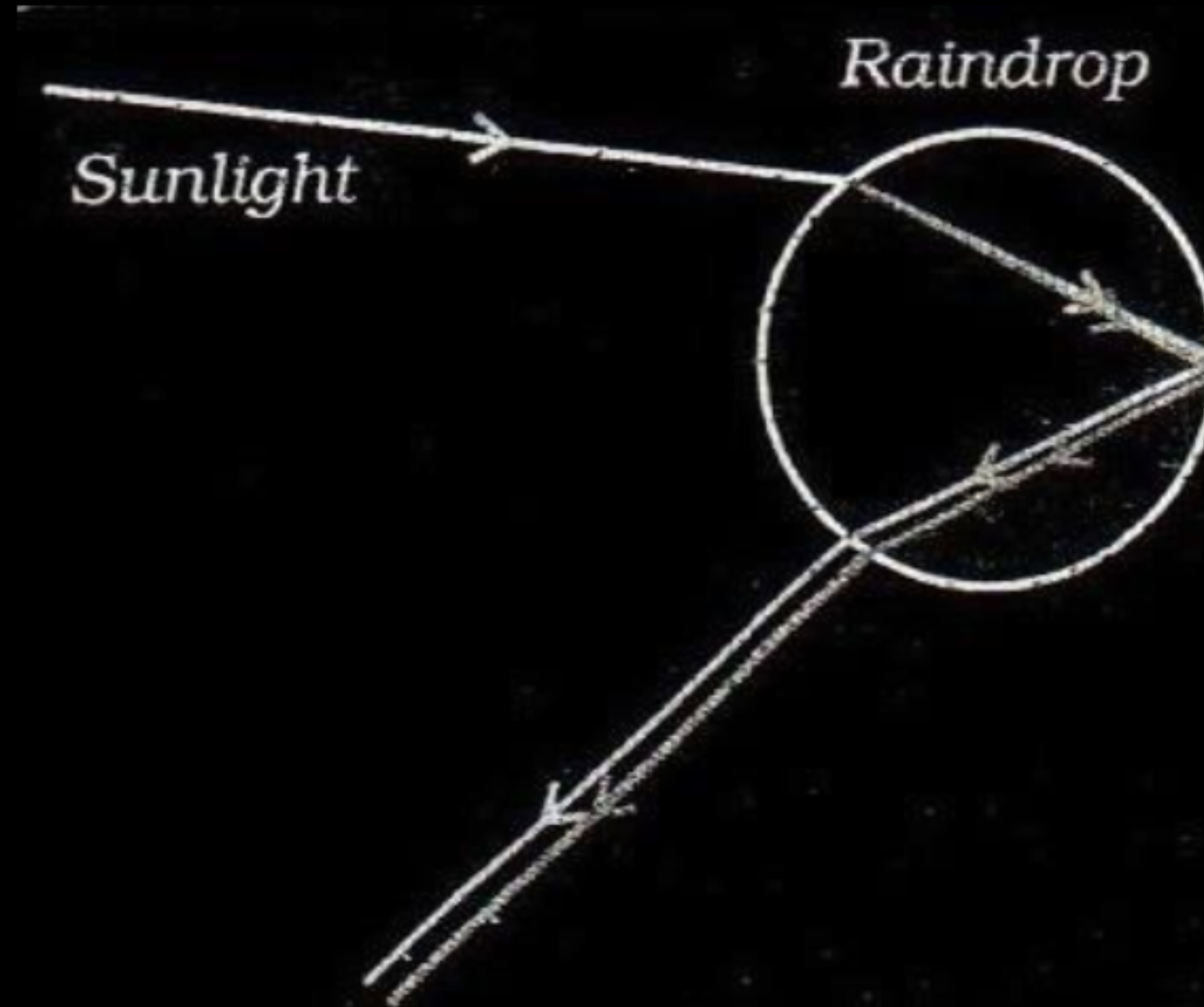
3



**31.** What is a rainbow ? Draw a labelled diagram to show its formation.

3

A rainbow is a natural spectrum appearing in the sky after a rain shower.



(Deduct half mark if arrows are not marked)

1

2



**13.** The white light entering a glass prism, gets split into its constituent colours. It is observed that :

- (A) Red light deviates the most.
- (B) Violet light deviates the least.
- (C) Yellow light deviates more than the blue light.
- (D) Green light deviates more than the orange light.



**13.** The white light entering a glass prism, gets split into its constituent colours. It is observed that :

- (A) Red light deviates the most.
- (B) Violet light deviates the least.
- (C) Yellow light deviates more than the blue light.
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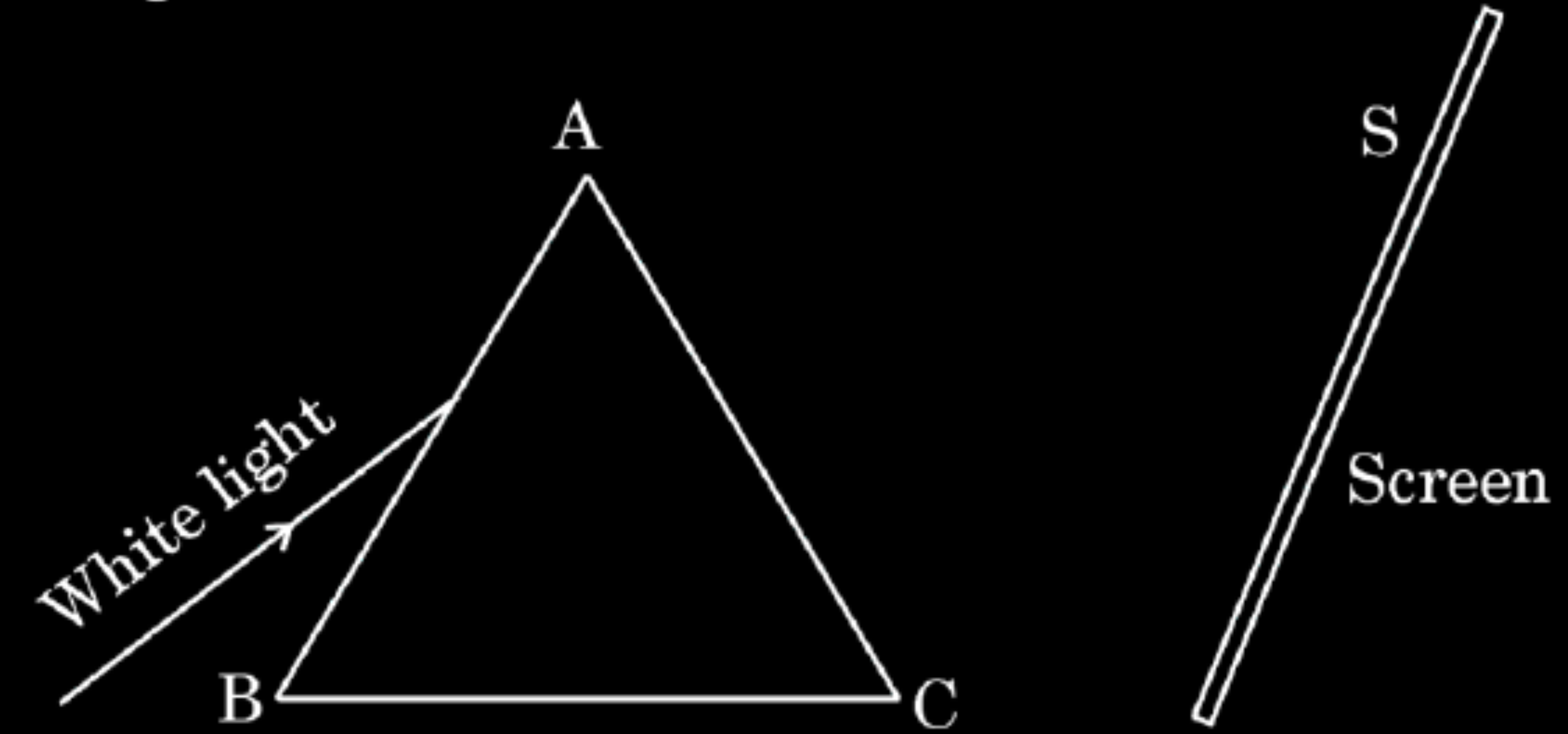
---

D/Green light deviates more than the orange light

---



39. A person allowed a narrow beam of white light from the sun to enter a dark room through a small aperture and placed a glass prism in its path in such a manner that the beam falls on the face AB of the prism as shown in the figure.



A screen S is placed on the other side of the prism, facing AC. On turning the prism slowly, a beautiful band of colours is obtained on the screen. It is the spectrum of sunlight.

- (a) Name the phenomenon due to which a prism splits the incident white light into a band of colours. 1
- (b) State the reason of getting a band of seven colours in the above case. 1
- (c) (i) Explain with the help of a labelled ray diagram, an experimental arrangement to show the recombination of the spectrum of white light. 2

**OR**

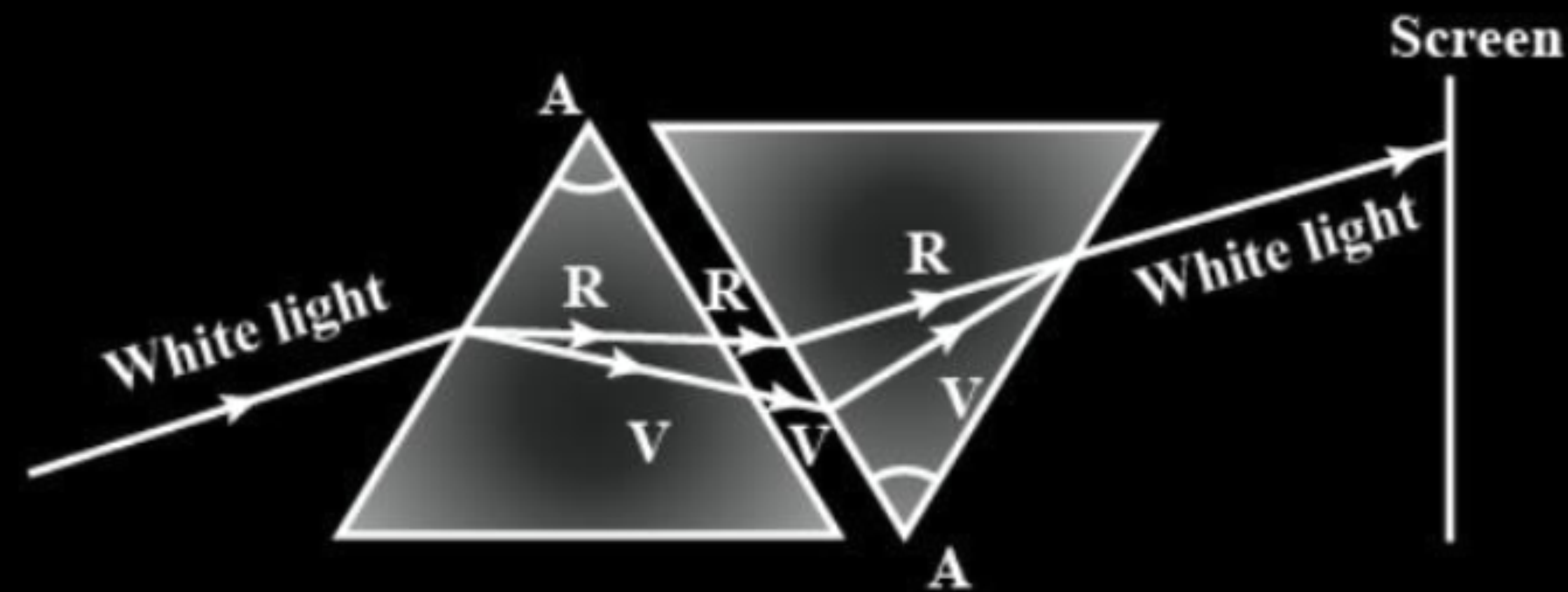
- (c) (ii) Draw a labelled ray diagram to show the formation of a rainbow. 2



- (a) Dispersion of light
- (b) Different colours of light bend through different angles with respect to the incident ray as they pass through a prism.
- (c) (i) Two identical prisms are placed in inverted position with respect to each other as shown. When spectrum produced by prism A is passed through the prism B, a beam of white light emerges from the other side of the prism B.

1

1

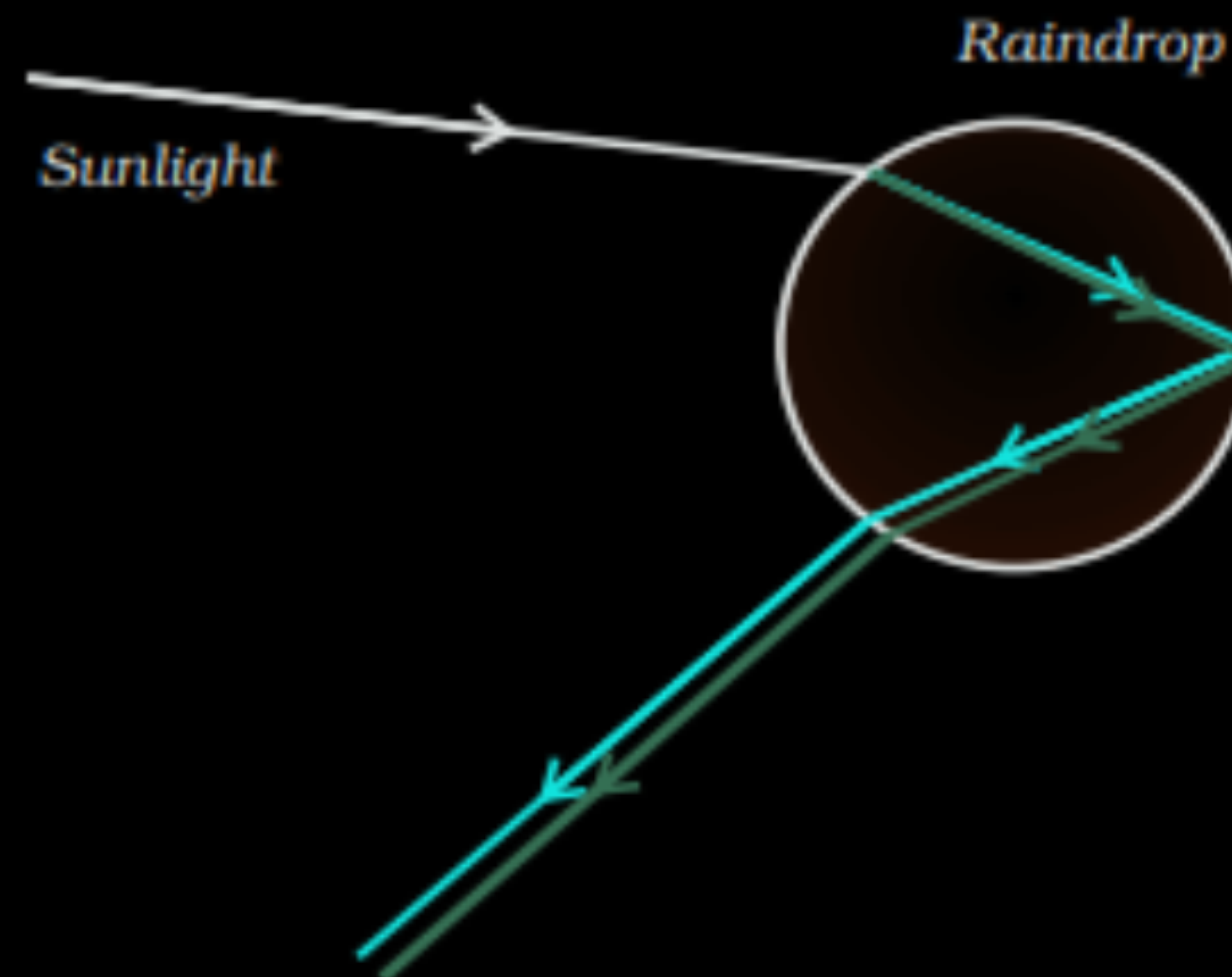


(award full marks even if only labelled ray diagram is given)

2

OR

(c)(ii)

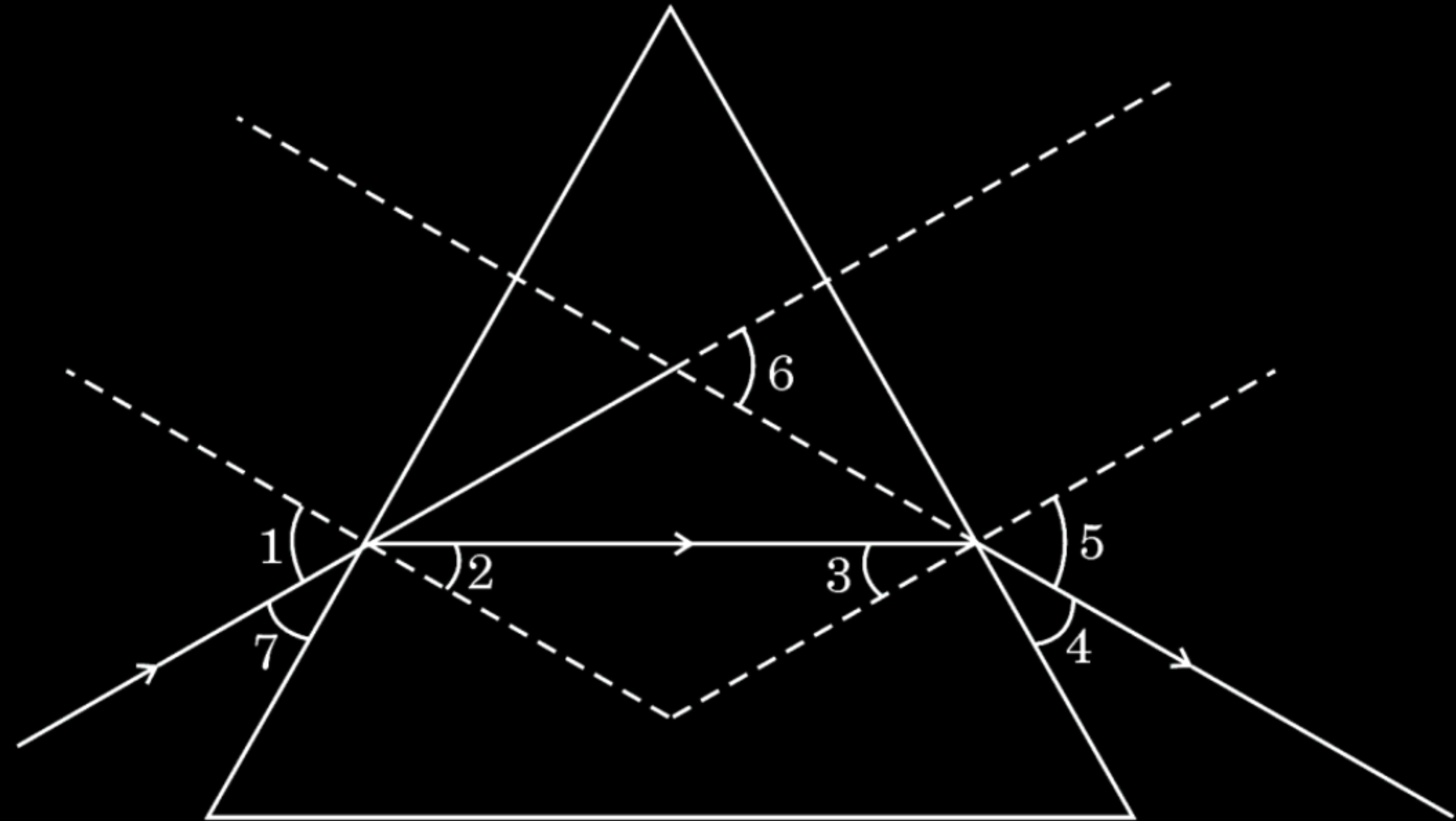


(deduct ½ marks if arrows are not marked)

2



13. In the given figure the angle of incidence and the angle of deviation respectively are :

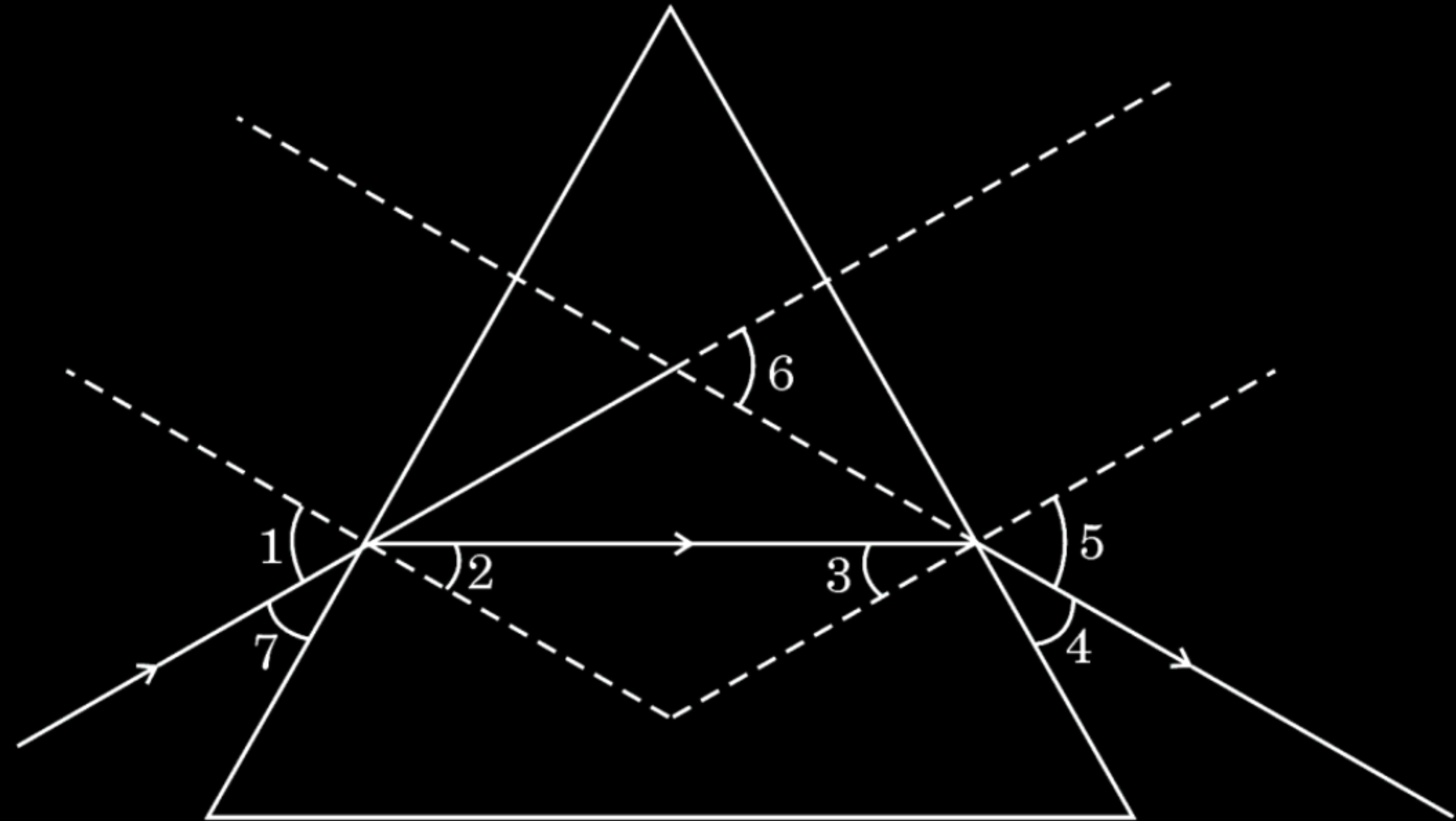


- (A) 1 and 5  
(C) 7 and 4

- (B) 7 and 6  
(D) 1 and 6



13. In the given figure the angle of incidence and the angle of deviation respectively are :



- (A) 1 and 5  
(C) 7 and 4

- (B) 7 and 6  
(D) 1 and 6

(D) / 1 and 6



**28.** When and where does a rainbow appear in the sky ? List the phenomena of light in proper sequence which occurs during the formation of a rainbow. Why can a rainbow not be seen from the surface of the moon ?



24. How will you use two identical glass prisms so that a narrow beam of white light incident on one prism emerges out of the second prism as white light ? Draw and label the ray diagram.

3

By placing second (identical) prism in an inverted position with respect to the first prism.

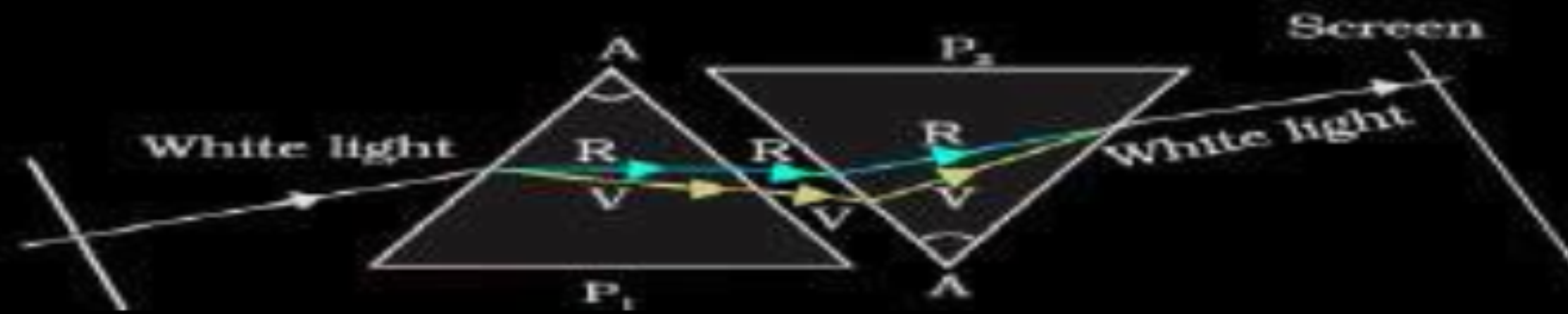


Diagram  
Labelling

1

$1\frac{1}{2}$   
 $\frac{1}{2}$



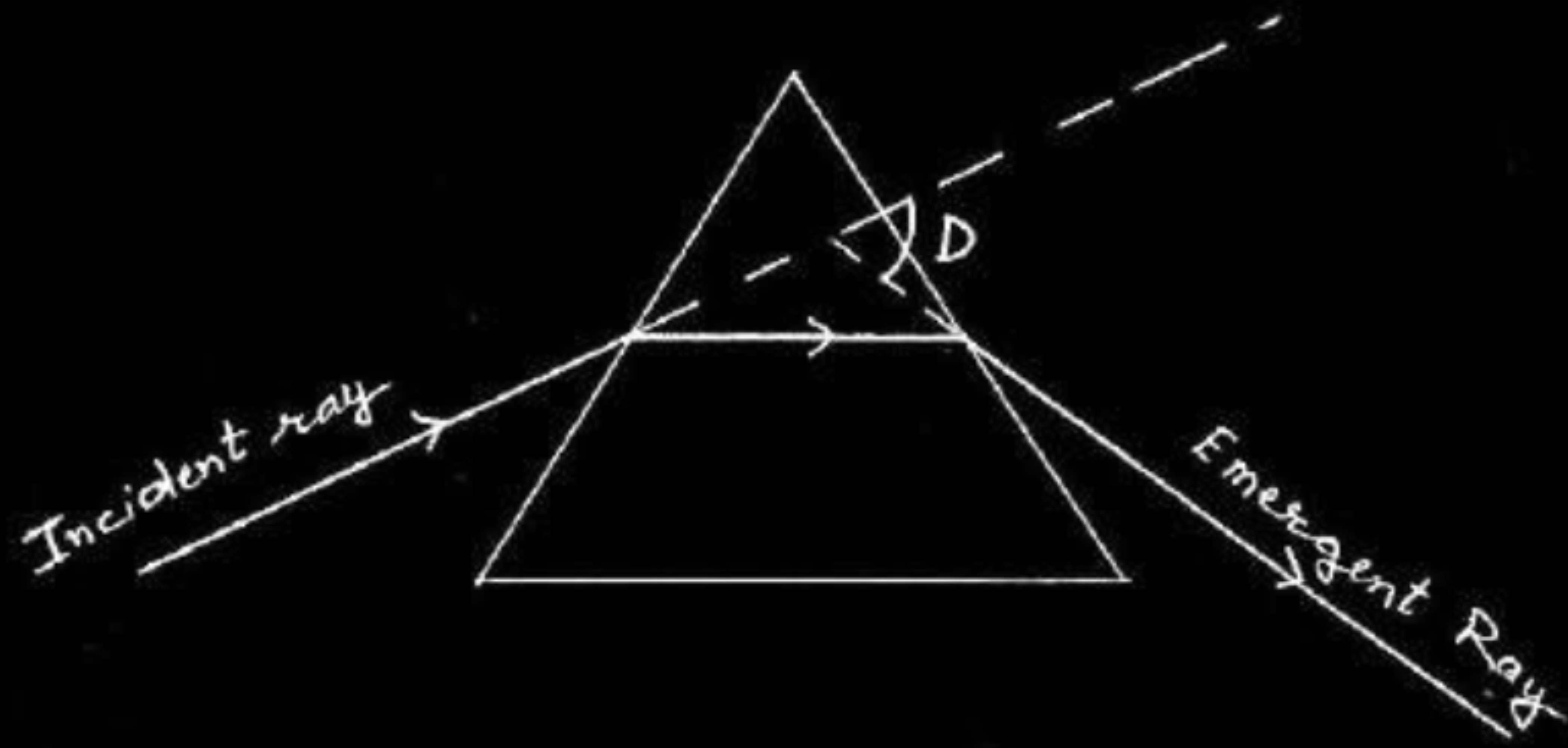
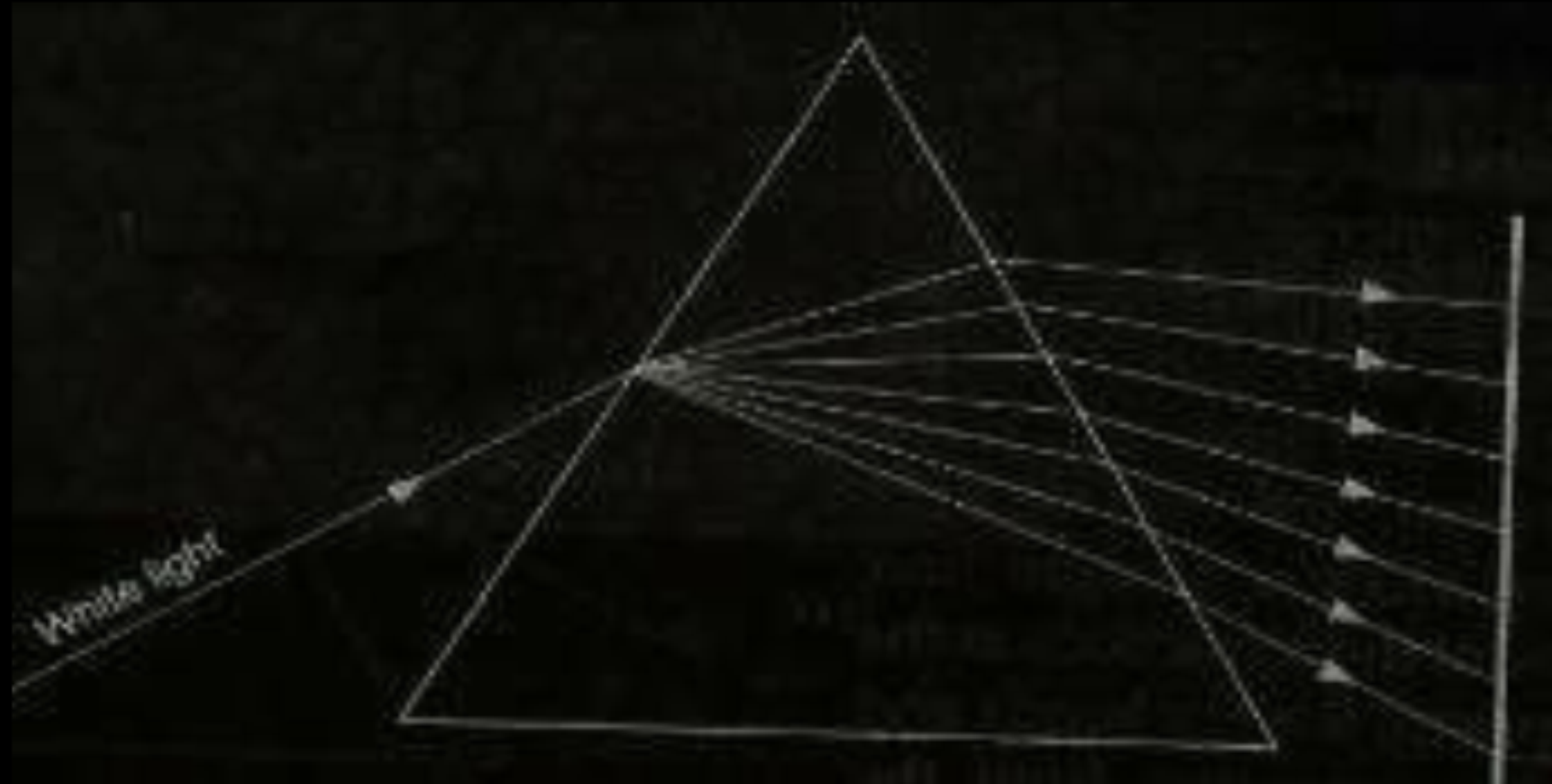
24. (a) With the help of labelled ray diagram show the path followed by a narrow beam of monochromatic light when it passes through a glass prism.
- (b) What would happen if this beam is replaced by a narrow beam of white light ?



24. (a) With the help of labelled ray diagram show the path followed by a narrow beam of monochromatic light when it passes through a glass prism.

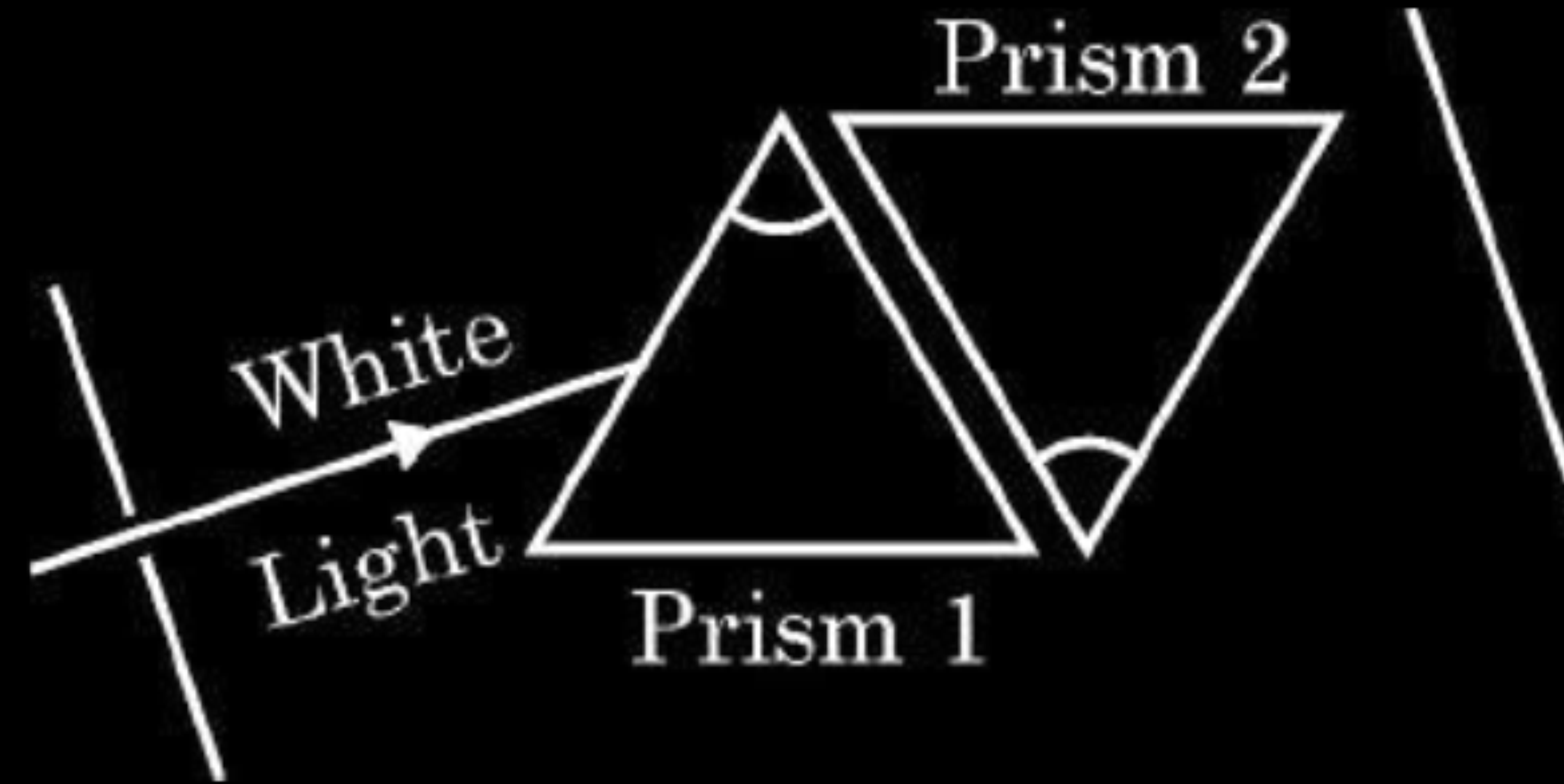
(b) What would happen if this beam is replaced by a narrow beam of white light ?

3

|                                                                                                                                     |                |                                                                                                                                                                                                                                                  |          |
|-------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <p>(a)</p>  <p>Path of the ray<br/>Labelling</p> | <p>1<br/>1</p> | <p>(b) Splitting into seven colours / Dispersion / VIBGYOR /</p>  <p><b>Note :</b> Marks may also be awarded if answer is given in the form of a figure.</p> | <p>1</p> |
|-------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

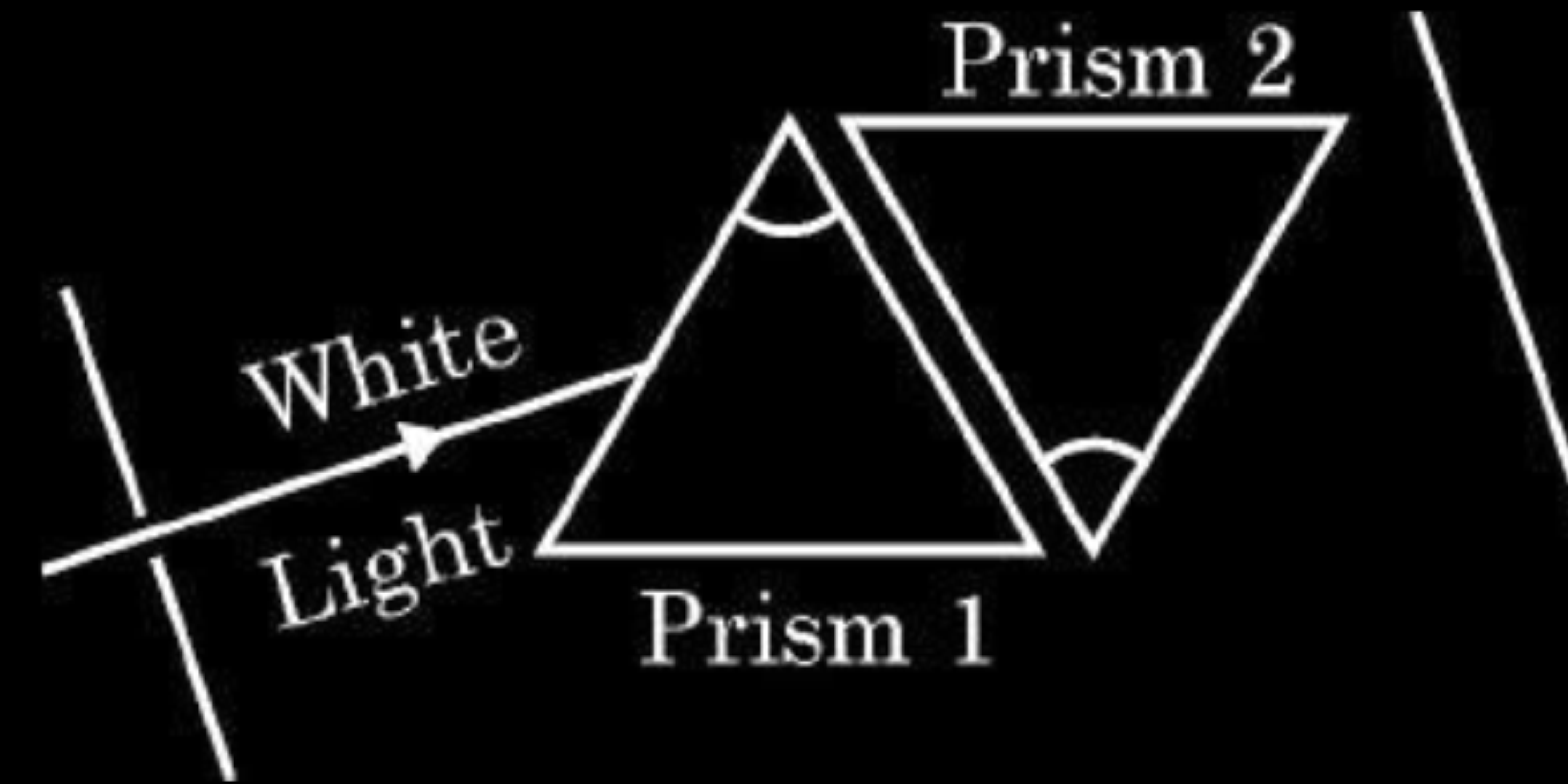


- (a) State the relation between colour of scattered light and size of the scattering particle.
- (b) The apparent position of an object, when seen through the hot air, fluctuates or wavers. State the basic cause of this observation.
- (c) Complete the path of white light when it passes through two identical prisms placed as shown :



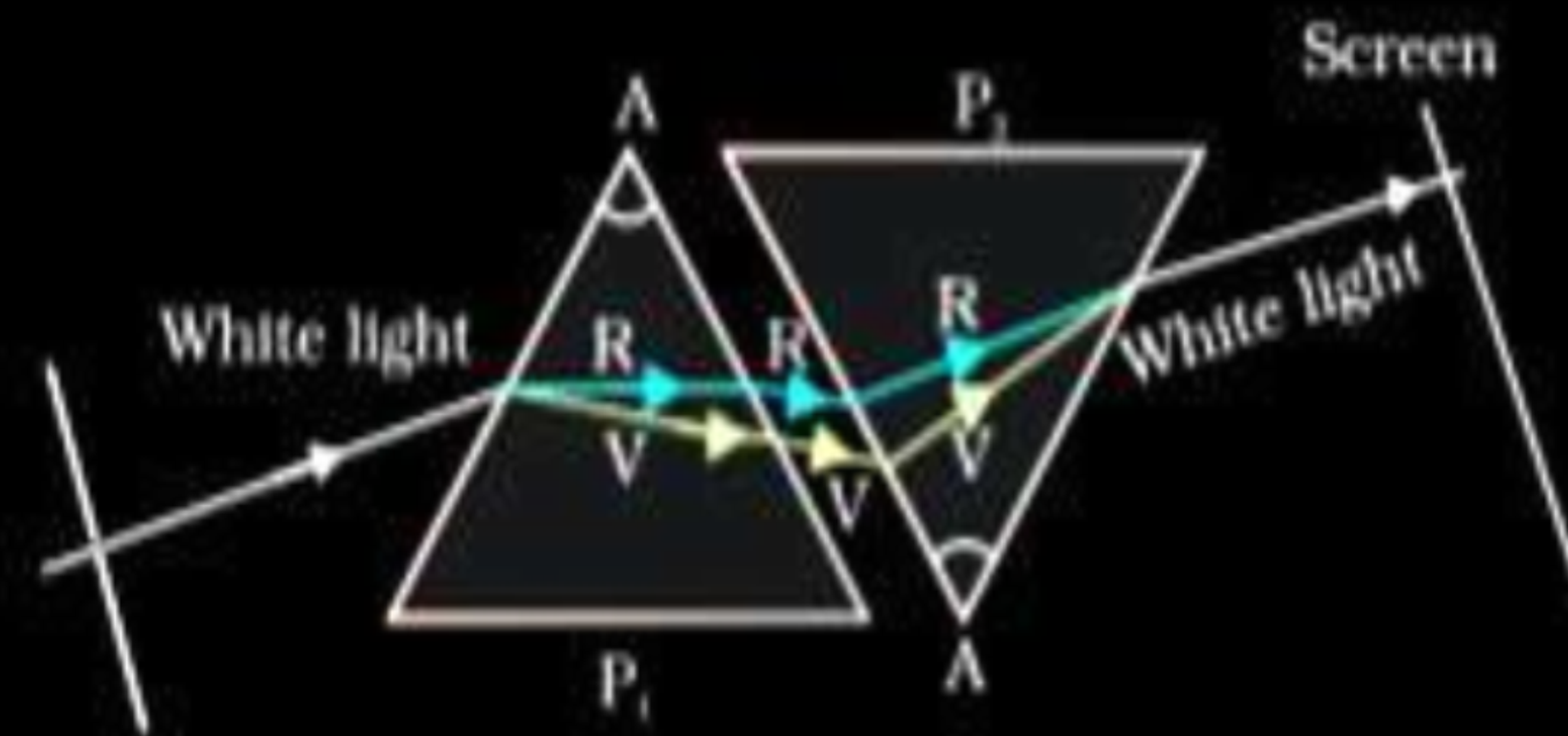


- (a) State the relation between colour of scattered light and size of the scattering particle.
- (b) The apparent position of an object, when seen through the hot air, fluctuates or wavers. State the basic cause of this observation.
- (c) Complete the path of white light when it passes through two identical prisms placed as shown :



3

- (a) Small size particles scatter shorter wavelength (violet) or large sized particles scatter larger wavelength (Red).
- (b) Due to variation in physical condition of hot air.
- (c) Diagram



(Splitting of white light is essential )

1



Differentiate between a glass slab and a glass prism. What happens when a narrow beam of (i) a monochromatic light, and (ii) white light passes through (a) glass slab and (b) glass prism?



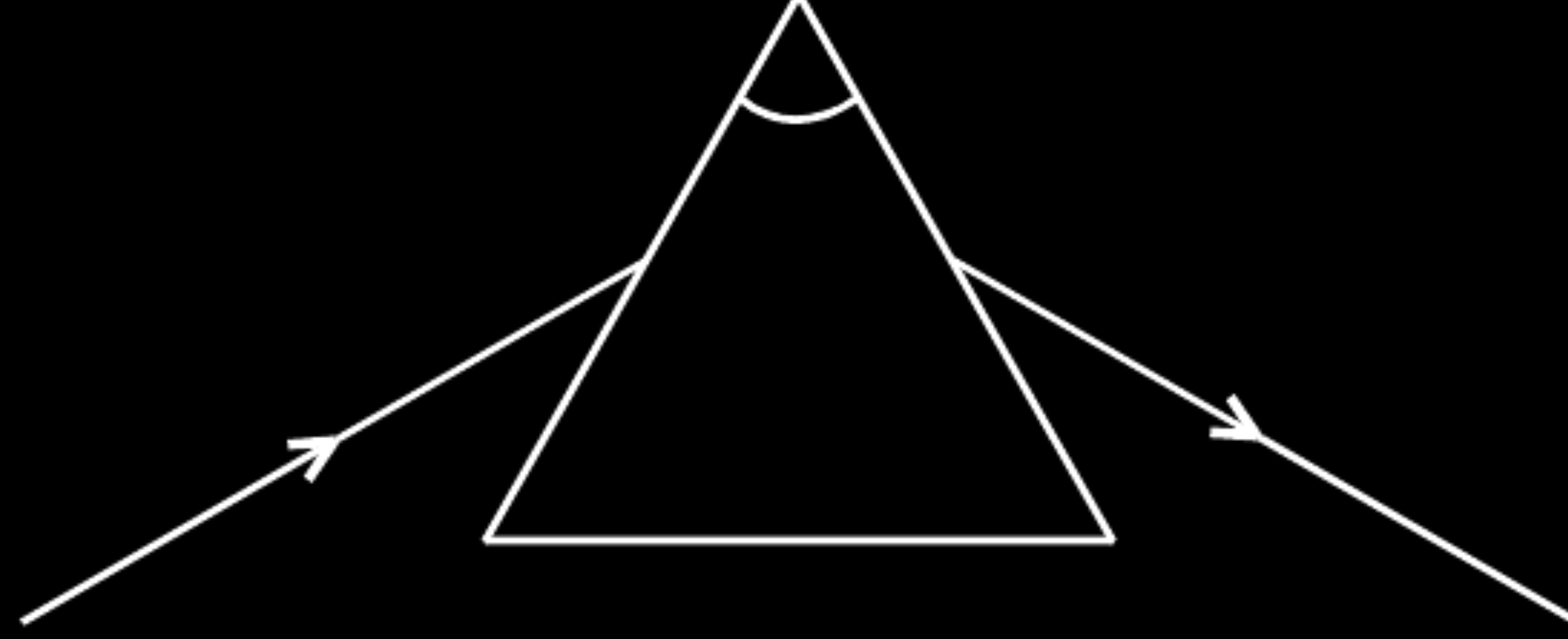
Differentiate between a glass slab and a glass prism. What happens when a narrow beam of (i) a monochromatic light, and (ii) white light passes through (a) glass slab and (b) glass prism?

3

|                                                                                                                                                     |               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <ul style="list-style-type: none"> <li>Prism has 2 inclined refracting surfaces whereas a glass slab has 2 parallel refracting surfaces.</li> </ul> | 1             |
| i) When monochromatic light passes through a glass slab it gets displaced laterally whereas in prism it gets angularly displaced.                   | $\frac{1}{2}$ |
| ii) When white light passes through a glass slab, it gets laterally displaced whereas in prism, dispersion takes place.                             | $\frac{1}{2}$ |

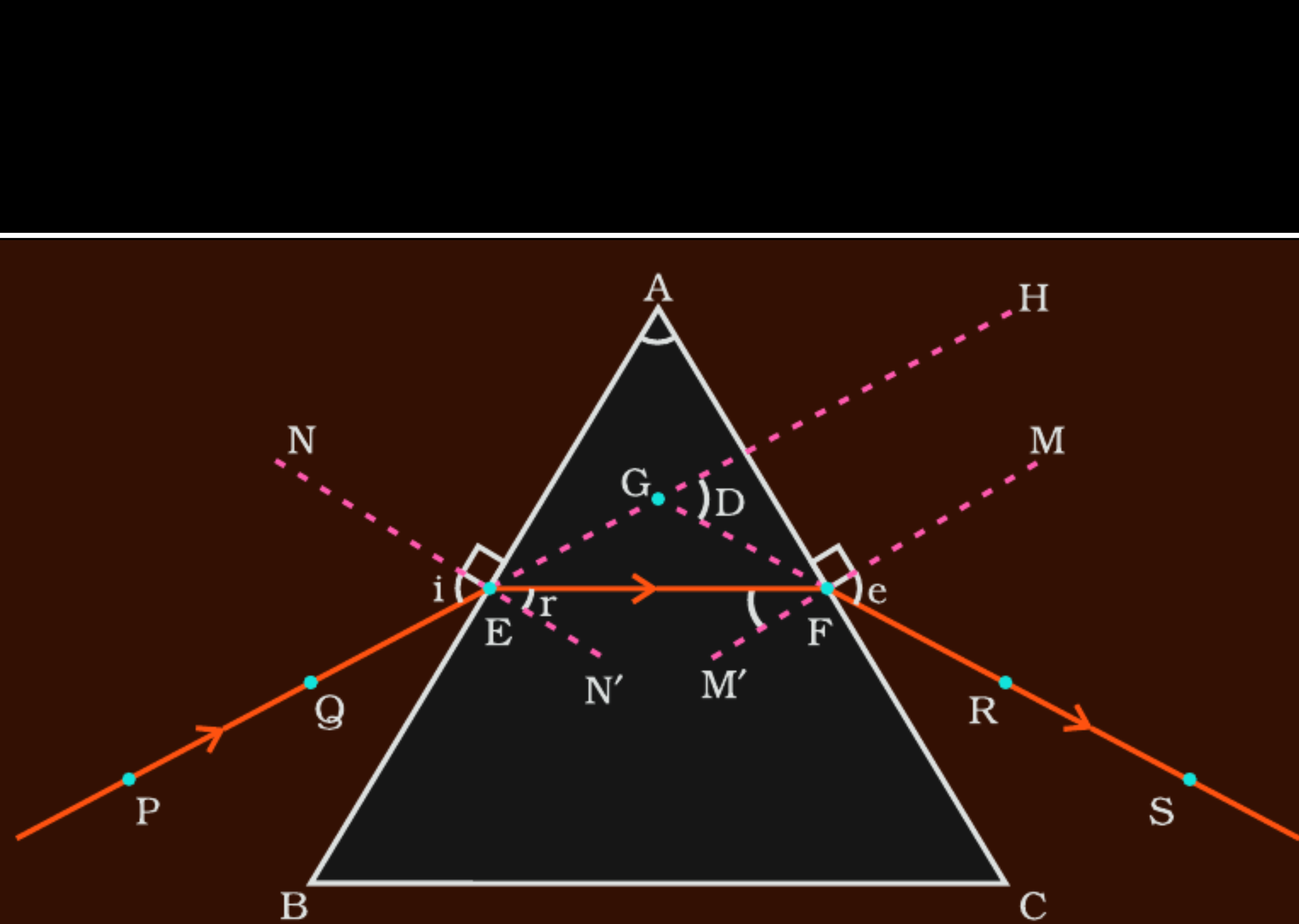


A student traces the path of a ray of light through a glass prism as shown in the diagram, but leaves it incomplete and unlabelled. Redraw and complete the diagram. Also label on it  $\angle i$ ,  $\angle e$ ,  $\angle r$  and  $\angle D$ .





A student traces the path of a ray of light through a glass prism as shown in the diagram, but leaves it incomplete and unlabelled. Redraw and complete the diagram. Also label on it  $\angle i$ ,  $\angle e$ ,  $\angle r$  and  $\angle D$ .



PE – Incident ray

EF – Refracted ray

FS – Emergent ray

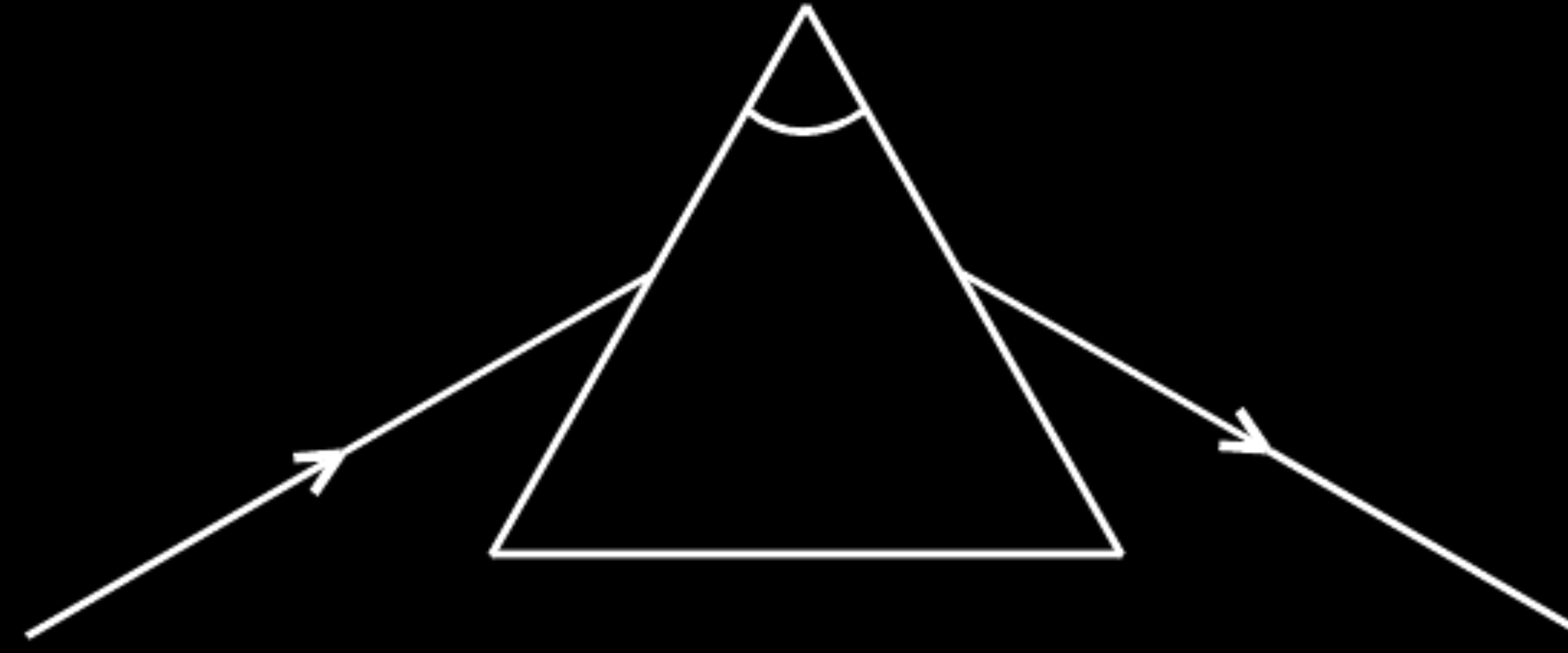
$\angle A$  – Angle of the prism

$\angle i$  – Angle of incidence

$\angle r$  – Angle of refraction

$\angle e$  – Angle of emergence

$\angle D$  – Angle of deviation



Labelling of  $\angle i + \angle e + \angle r$  &  $\angle D$

$\frac{1}{2} \times 4$



What happens to a beam of white light when it gets refracted through a glass prism ? Which colour deviates the most and the least after refraction through a prism ? What is likely to happen if a second identical prism is placed in an inverted position with respect to the first prism ? Justify your answer.



What happens to a beam of white light when it gets refracted through a glass prism ? Which colour deviates the most and the least after refraction through a prism ? What is likely to happen if a second identical prism is placed in an inverted position with respect to the first prism ? Justify your answer.

|                                                                                                    |               |
|----------------------------------------------------------------------------------------------------|---------------|
| The white light splits into seven colours when it gets refracted through the glass prism (VIBGYOR) | 1             |
| The colour deviates most – Violet                                                                  | $\frac{1}{2}$ |
| The colour deviates least – Red                                                                    | $\frac{1}{2}$ |
| Colours disappear and again white light obtained.                                                  | 1             |



A glass prism is able to produce a spectrum when white light passes through it but a rectangular block of same transparent glass does not produce any spectrum. Why ?



A glass prism is able to produce a spectrum when white light passes through it but a rectangular block of same transparent glass does not produce any spectrum. Why ?

|                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The extent of bending of a light ray at core opposite parallel faces of the rectangular glass slab is equal. Therefore, the emergent ray is parallel to the incident ray. So it cannot produce any spectrum. |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



What is the cause of dispersion of white light through a glass prism ? Draw a ray diagram to show the path of light when two identical glass prisms are arranged together in inverted position with respect to each other and a narrow beam of white light is allowed to fall obliquely on one of the faces of the prisms.

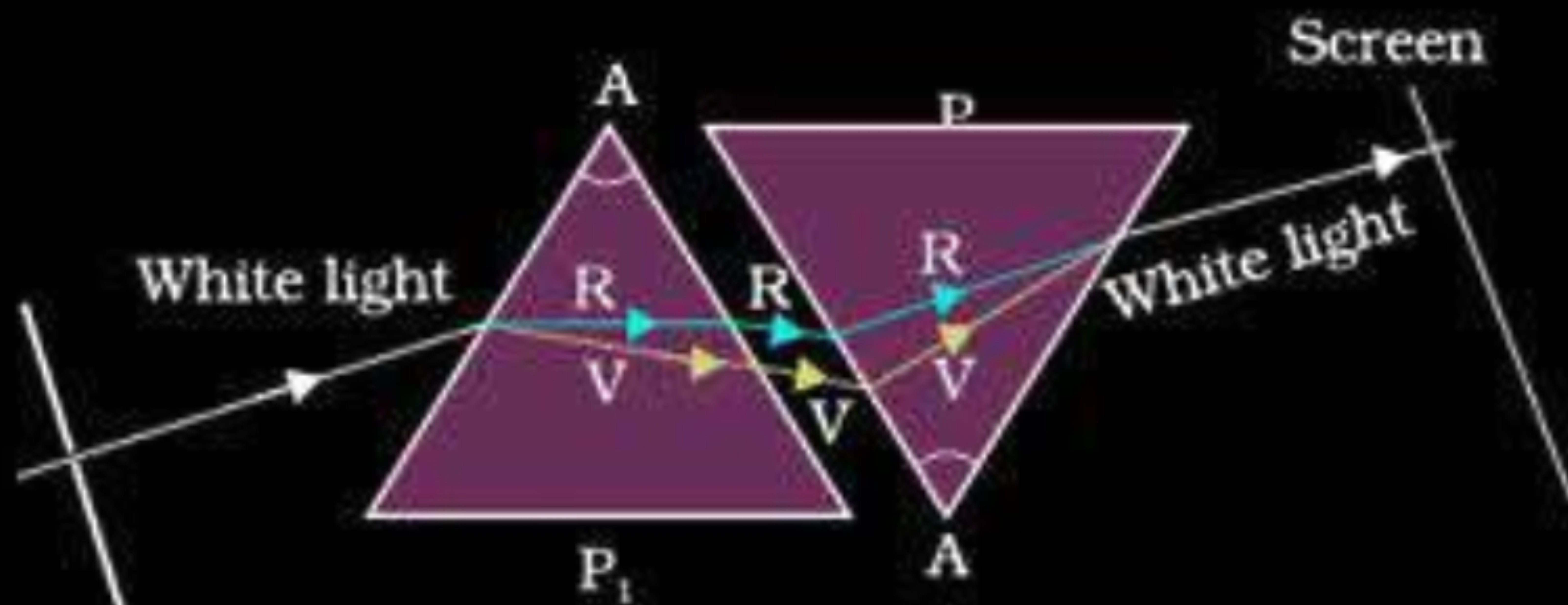


What is the cause of dispersion of white light through a glass prism ? Draw a ray diagram to show the path of light when two identical glass prisms are arranged together in inverted position with respect to each other and a narrow beam of white light is allowed to fall obliquely on one of the faces of the prisms.

- Cause of dispersion:  
 (i) Shape of prism  
 (ii) Different colours bend with different angles (different colour has different refractive index or different speed)

diagram

Arrow& labelling



1

1

1

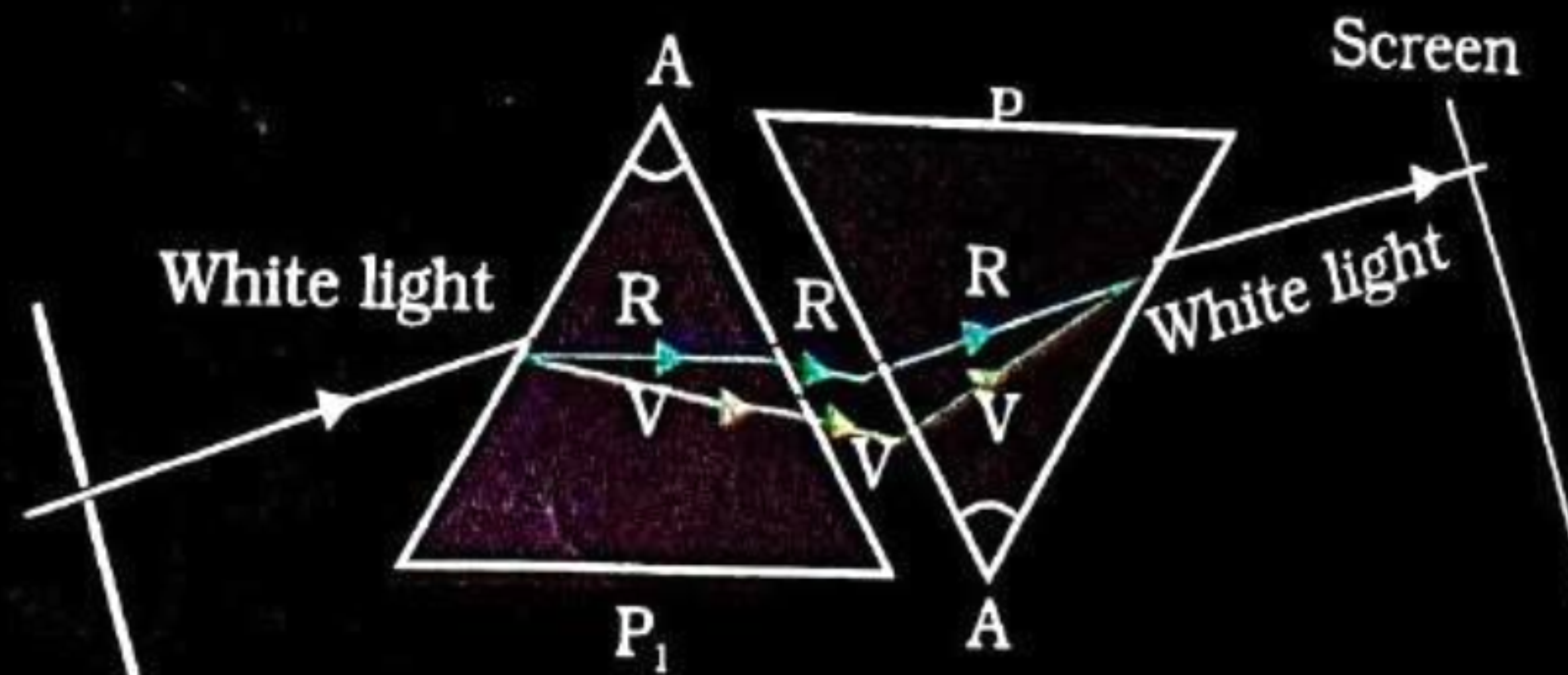


State the cause of dispersion of white light by a glass prism. How did Newton, using two identical glass prisms, show that white light is made of seven colours ? Draw a ray diagram to show the path of a narrow beam of white light, through a combination of two identical prisms arranged together in inverted position with respect to each other, when it is allowed to fall obliquely on one of the faces of the first prism of the combination.



State the cause of dispersion of white light by a glass prism. How did Newton, using two identical glass prisms, show that white light is made of seven colours ? Draw a ray diagram to show the path of a narrow beam of white light, through a combination of two identical prisms arranged together in inverted position with respect to each other, when it is allowed to fall obliquely on one of the faces of the first prism of the combination.

Different colours of light bend through different angles with respect to the incident ray / different speed of different colours of light in glass / different values of refractive index of glass for different colours of light.



Direction of ray & labelling

1

1

$\frac{1}{2} + \frac{1}{2}$



Describe an activity to show that the colours of white light splitted by a glass prism can be recombined to get white light by another identical glass prism. Also draw ray diagram to show the recombination of the spectrum of white light.



Describe an activity to show that the colours of white light splitted by a glass prism can be recombined to get white light by another identical glass prism. Also draw ray diagram to show the recombination of the spectrum of white light.

Description of activity- When a glass prism is used to obtain a spectrum of sunlight, a second identical prism in an inverted position with respect to the first position will allow all the colours of spectrum to recombine .Thus a beam of white light will emerge from the other side of the second prism.

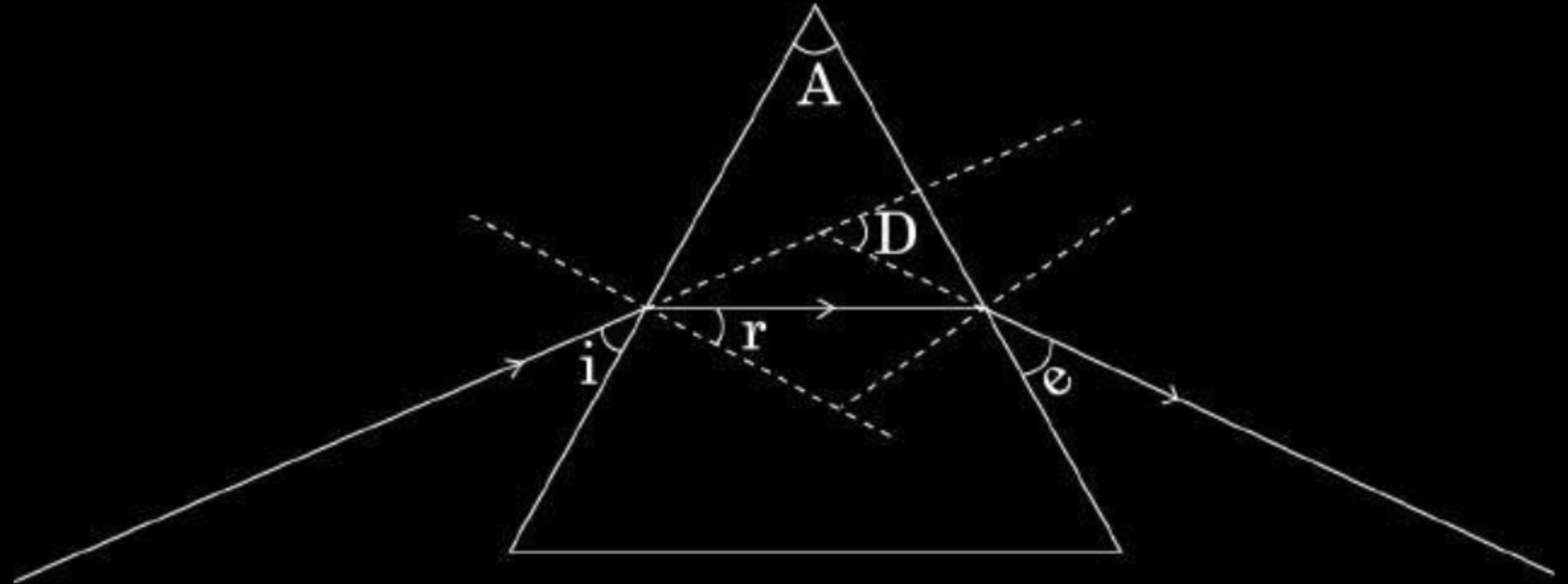
1 ½



1 ½



In the following ray diagram the correctly marked angle are :



(a)  $\angle i$  and  $\angle e$

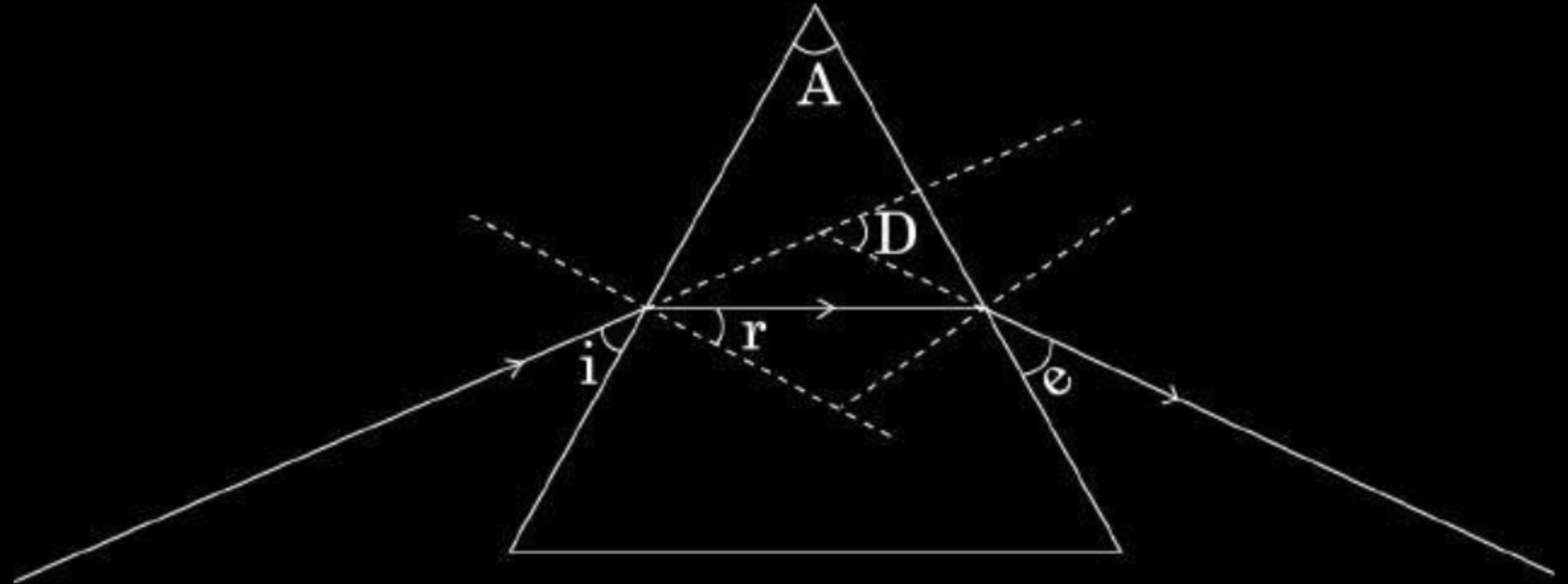
(b)  $\angle A$  and  $\angle D$

(c)  $\angle i$ ,  $\angle e$  and  $\angle D$

(d)  $\angle r$ ,  $\angle A$  and  $\angle D$



In the following ray diagram the correctly marked angle are :



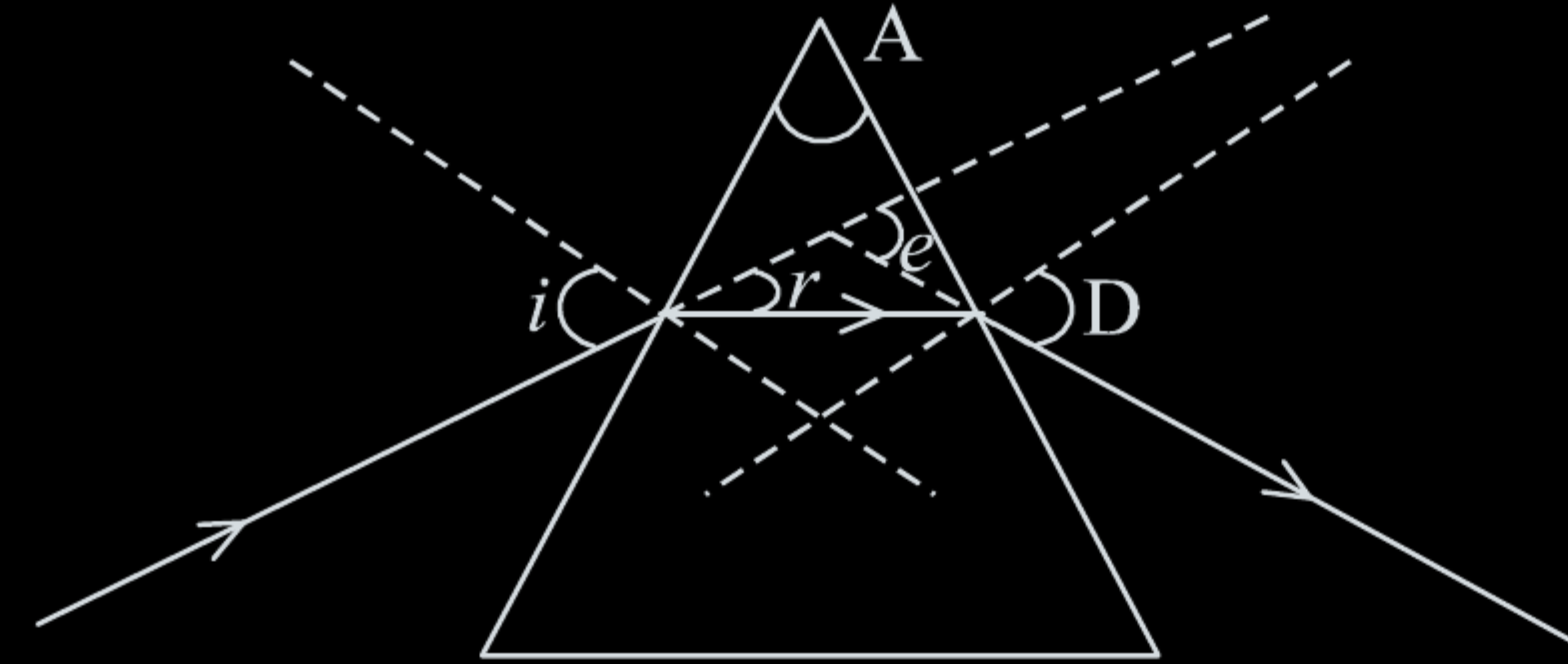
- (a)  $\angle i$  and  $\angle e$   
(c)  $\angle i$ ,  $\angle e$  and  $\angle D$

(b)  $\angle A$  and  $\angle D$

(d)  $\angle r$ ,  $\angle A$  and  $\angle D$



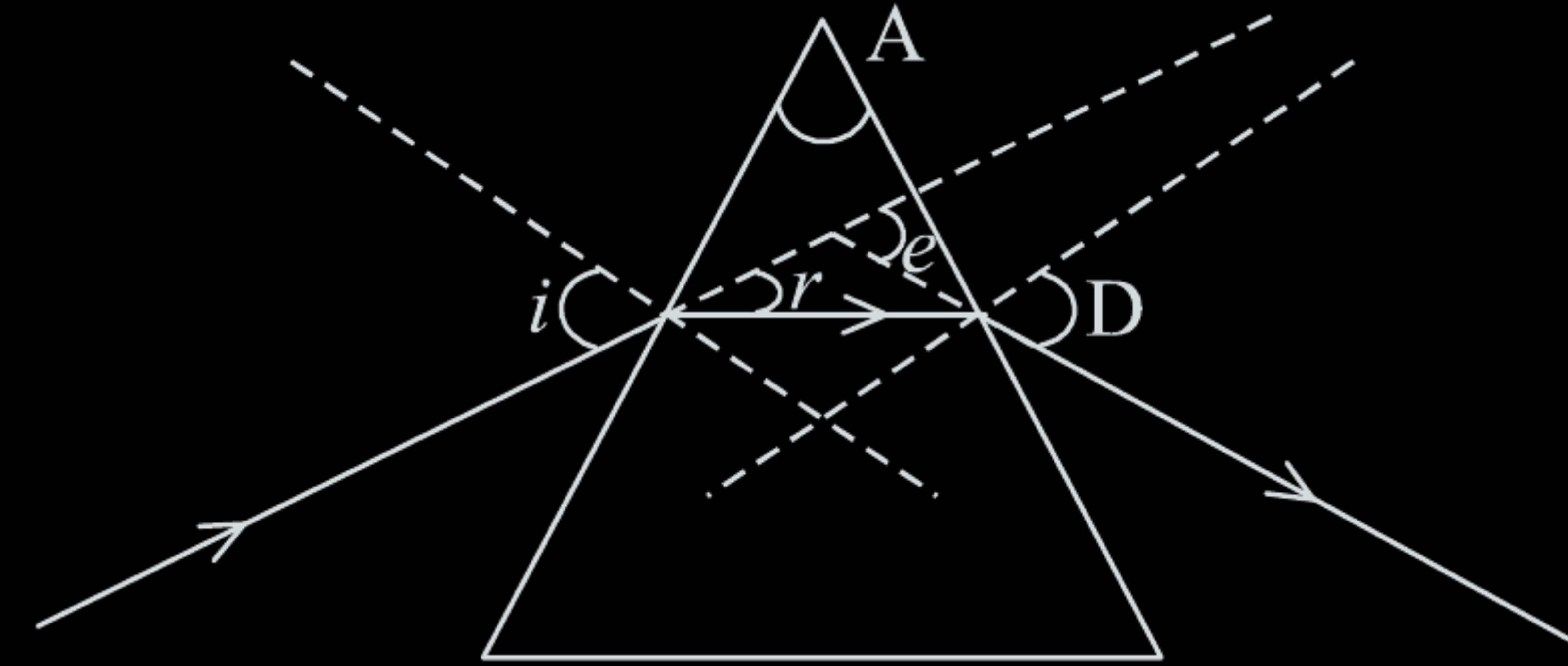
Study the following figure in which a student has marked the angle of incidence ( $\angle i$ ), angle of refraction ( $\angle r$ ), angle of emergence ( $\angle e$ ), angle of prism ( $\angle A$ ) and the angle of deviation ( $\angle D$ ). The correctly marked angles are :



- (a)  $\angle A$  and  $\angle i$
- (b)  $\angle A$ ,  $\angle i$  and  $\angle r$
- (c)  $\angle A$ ,  $\angle i$ ,  $\angle e$  and  $\angle D$
- (d)  $\angle A$ ,  $\angle i$ ,  $\angle r$  and  $\angle D$



Study the following figure in which a student has marked the angle of incidence ( $\angle i$ ), angle of refraction ( $\angle r$ ), angle of emergence ( $\angle e$ ), angle of prism ( $\angle A$ ) and the angle of deviation ( $\angle D$ ). The correctly marked angles are :



- (a)  $\angle A$  and  $\angle i$
- (b)  $\angle A$ ,  $\angle i$  and  $\angle r$
- (c)  $\angle A$ ,  $\angle i$ ,  $\angle e$  and  $\angle D$
- (d)  $\angle A$ ,  $\angle i$ ,  $\angle r$  and  $\angle D$